ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

- Discrete-time signals & difference equations (Ref pp. 9-19 and Ref pp. 43-59)
- Time and frequency domain analysis (Ref pp. 231-247 and Ref pp. 273-275)
- Sampling theorem & aliasing (Ref pp. 21-31 and Ref pp.396-413)
- A/D conversion & quantisation errors (Ref pp. 19-20 , Ref pp.31-37 and Ref pp.413-425)

Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

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Term 3, 2026

Digital Signal Processing (DSP) Fundamentals

What is DSP and where it is used?

- **Digital Signal Processing (DSP)** is the use of mathematical algorithms to process digital signals—to analyse, modify, extract information from, or improve them for a particular application.
- DSP developed rapidly from the 1970s and became widely used in the 1990s with the growth of digital audio, telecommunications and digital processors.
- Today, DSP is at the heart of many technologies we use every day:
  - **Audio & Speech:** music, smartphones, smart speakers, speech recognition
  - **Biomedical:** ECG analysis, medical monitoring and imaging
  - **Hearing:** digital hearing aids and hearing assessment
  - **Communications:** mobile phones, Wi-Fi and digital communications
  - **Radar & Sonar:** detection, tracking and navigation
  - **Control & Automotive:** sensing, control and autonomous systems

Digital Hearing Aid



Example DSP Applications

COMMUNICATIONS



- Echo Cancellation
- Digital PBXs
- Line Repeaters
- Modems
- Global Positioning
- Sound/Modem/Fax Cards
- Cellular Phones
- Speaker Phones
- Video Conferencing
- ATMs

VOICE / SPEECH



- Speech Recognition
- Speech Processing / Vocoding
- Speech Enhancement
- Text-to-Speech
- Voice Mail

PRO-AUDIO



- AV Editing
- Digital Mixers
- Home Theater
- Pro Audio

INSTRUMENTATION



- Spectrum Analyzers
- Seismic Processors
- Digital Oscilloscopes
- Mass Spectrometers

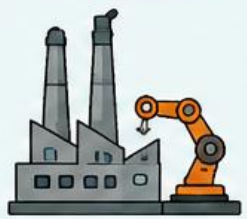
DSP
Digital Signal Processing

CONSUMER



- Radar Detectors
- Power Tools
- Digital Audio / TV
- Music Synthesizers
- Toys / Games
- Answering Machines
- Digital Speakers

INDUSTRIAL / CONTROL



- Robotics
- Numeric Control
- Power Line Monitors
- Motor / Servo Control

MEDICAL



- Patient Monitoring
- Ultrasound Equipment
- Diagnostic Tools
- Fetal Monitors
- Life Support Systems
- Image Enhancement

MILITARY

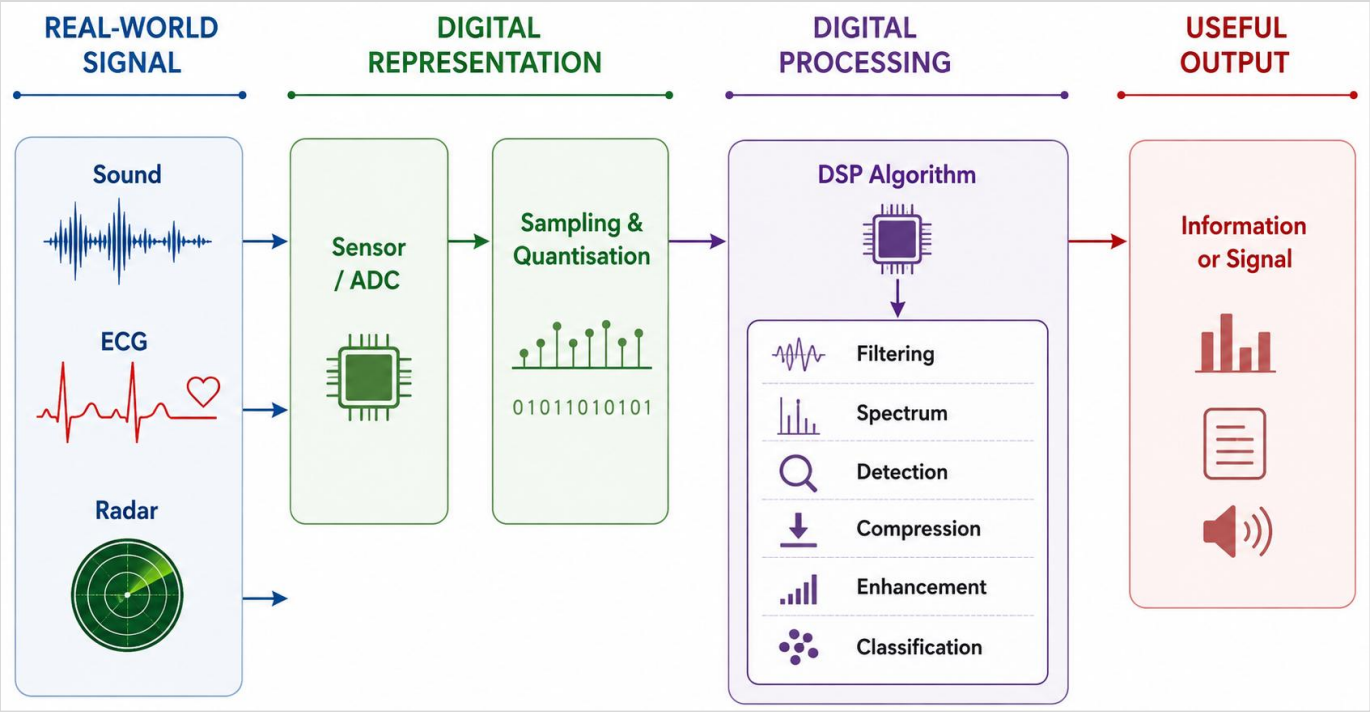


- Secure Communications
- Sonar Processing
- Image Processing
- Radar Processing
- Navigation, Guidance

Digital Signal Processing (DSP) Fundamentals

What Does DSP Actually Do?

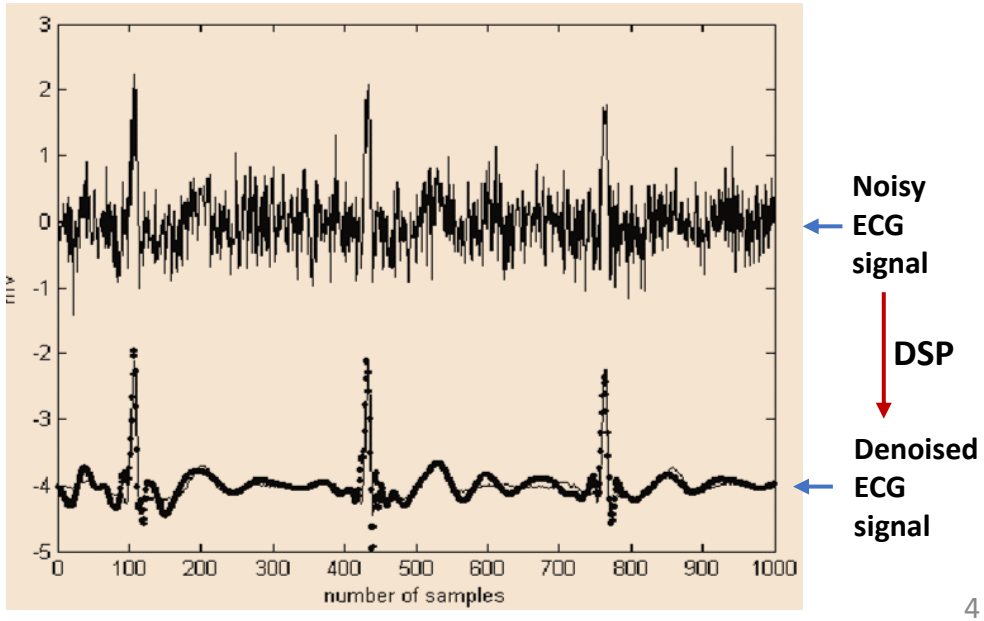
- DSP takes real-world signals (see diagram below), such as sound, ECG or radar signals, and converts them into digital form.
- The digital signal is then processed using DSP algorithms to filter noise, analyse its spectrum, detect information, compress data, enhance the signal or classify it.
- The result is a useful signal or useful information for a particular application.



Advantages of DSP

Why Digital Signal Processing?

- High accuracy and reliability
- Reproducible performance
- Flexible and programmable
- No analogue component drift
- Easy storage and transmission
- Powerful noise reduction and filtering
- Low-cost implementation



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Discrete-time signals

✓ Signals can be represented as either continuous-time or discrete-time signals.

Continuous - time signals

A continuous-time signal, $x(t)$, is defined for every value of the continuous independent variable t, where $-\infty < t < \infty$

Discrete - time signals

- Discrete-time signals are defined only at discrete values of time and consequently the independent variable n takes integer values .

  $(n = \ldots -2, -1, 0, 1, 2, \ldots)$

- A discrete- time signal is therefore a sequence of numbers represented by $x[n]$.

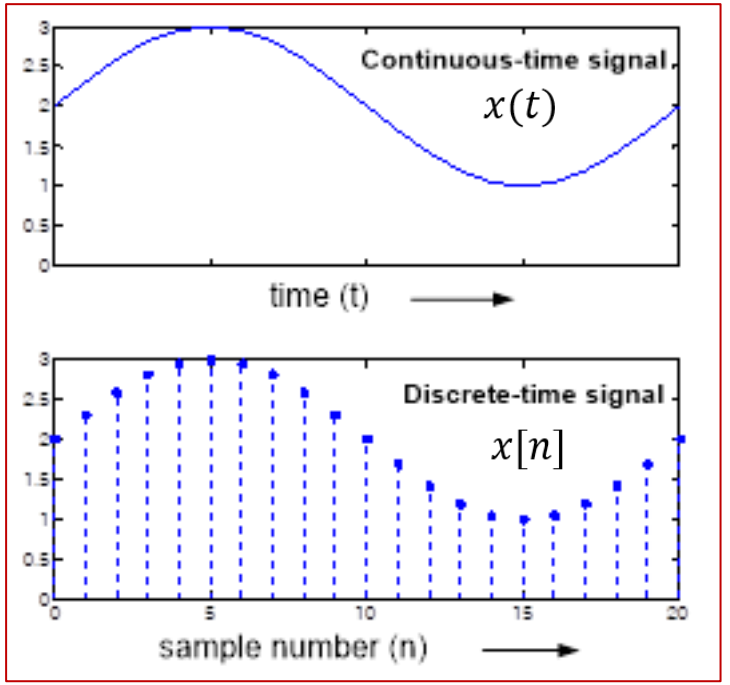


Figure 1: An example of continuous-time and discrete-time signals

Examples

- A person's body temperature varies continuously with time and can be represented as a **continuous-time signal**.
- A stock price recorded once every day is **a discrete-time signal.**
- Voltages and currents are **continuous-time signals.** When measured or sampled at discrete time intervals, they become **discrete-time signals.**

Discrete-time signals: Sampling

Sampling

A discrete-time signal $x[n]$ can be obtained by sampling a continuous-time signal $x(t)$ at equally spaced time intervals:

$$x[n] = x(t)\big|_{t=nT} = x(nT)$$

Square brackets [ ] → Discrete time signals $x[n]$

Round Brackets ( ) → Continuous signals $x(t)$

T is the **sampling period** (or sampling interval), and $f_s = 1/T$ is the sampling frequency.

$$x[n] = \{\ 3.5,\ \underset{n=-1}{4},\ 3.25,\ \underset{n=0}{2},\ 2.5,\ \underset{n=2}{3.0},\ \underset{n=4}{}\ \}$$

Important: *x[n]* is defined only for integer values of *n*. For example, x[1.5] is **undefined**, not zero. *x[n]* is simply undefined for non-integer values of *n*.

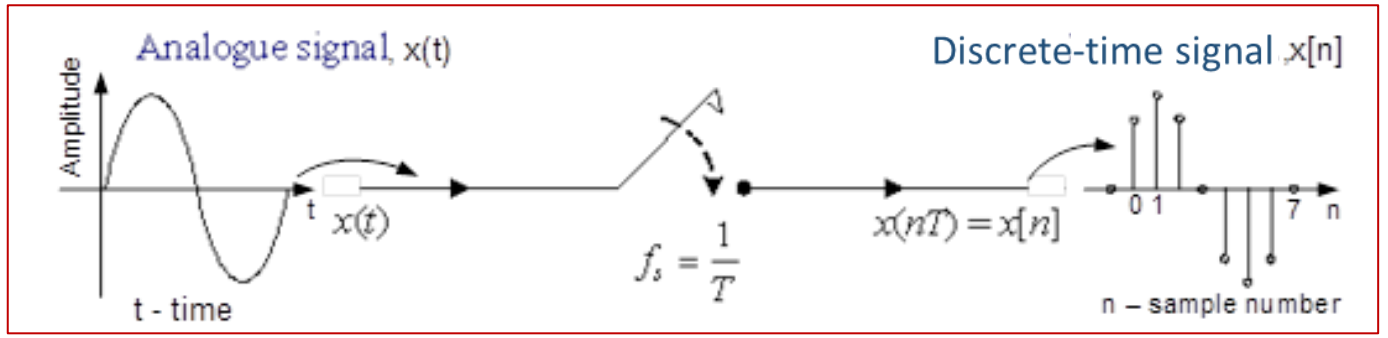


Figure 2: Obtaining a discrete-time signal by sampling a continuous-time signal.

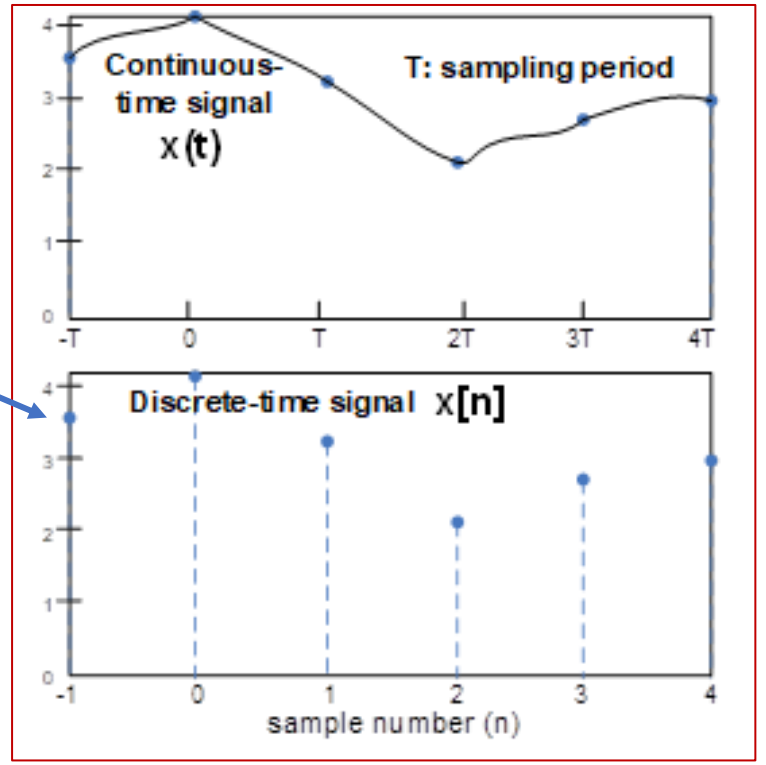
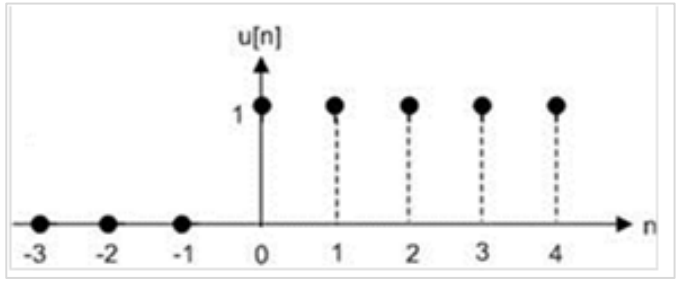


Figure 3: Continuous-time signal and its sampled discrete-time representation.

Discrete-time Signals: Unit step, Unit impulse & Exponential Signals

Unit step function

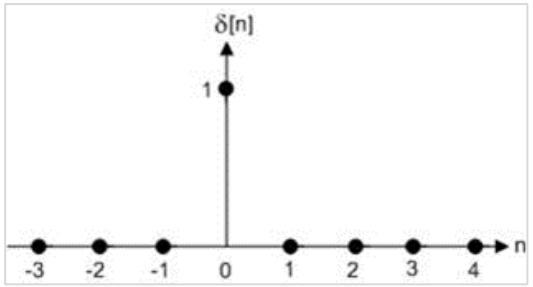
$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



Unit Impulse Function

- The discrete-time unit impulse function $\delta[n]$ is defined as

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$



- It is also called unit sample.
- $\delta[n] = u[n] - u[n-1]$

Discreate – time exponential signals

- A real discrete-time exponential signal can be written as

$$x[n] = c^{\ n}$$

where c and α are constants.

- For $c > 0$ and $\alpha > 0$:

  α >1: growing exponential

  0 < α <1: decaying exponential

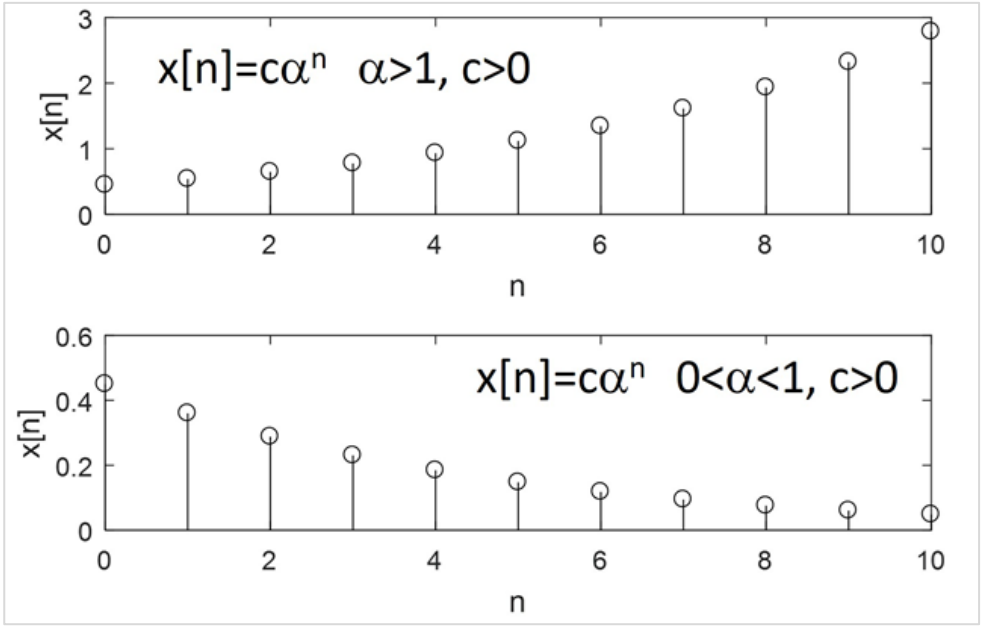


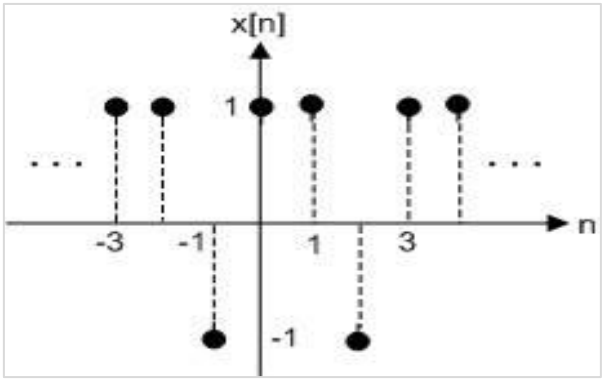
Figure 4: Examples of discrete-time exponential signals

Discrete-time signals : Periodic signals

Periodic Signals

- A discrete-time signal *x[n]* is periodic with period *N*, where *N* is a positive integer, if for all values of *n*.

$$x[n] = x[n+N]$$



x[n] with Period, *N* = 3 samples

Discrete-time sinusoidal signals

$$x(t) = A\sin(\omega_a t) = A\sin(2\pi f_a t)$$

- A continuous-time sinusoidal signal is given by:
- discrete - time sinusoidal signal may be expressed as: $x[n] = x(t)|_{t = nT} = x(nT)$

$$x[n] = A\sin(n\boldsymbol{\omega_a}\mathrm{T}) = A\sin\left(2\pi\frac{\boldsymbol{f_a}}{\boldsymbol{f_s}}\mathrm{n}\right) = Asin(n\theta)$$

$Sampling\ frequency\ (f_s) = \frac{1}{T}$ Hz; Sample number = n; f_a: Analogue frequency

$Digital\ frequency\ (\theta) = \boldsymbol{2\pi\frac{f_a}{f_s} = \omega_a T}$ radians ; where $\boldsymbol{-\pi \le \theta \le \pi}$ (Principal frequency range)

Discrete-time signals : Periodic signals

- A discrete-time signal is said to be periodic with a period length *N*, if *N* is the smallest positive integer for which

$$x[n+N]=x[n]$$
$$\therefore A\sin\big((n+N)\theta\big)=A\sin(n\theta)$$

which can only be satisfied for all *n* if, $N\theta=2\pi k$ (where *k* is an arbitrary integer); $\therefore N=\frac{2\pi k}{\theta}=\frac{2\pi k}{2\pi\frac{f_a}{f_s}}$

$$N=\frac{f_s}{f_a}k \qquad N\frac{f_a}{f_s}=k$$

- For example, f_a = 1000 Hz and f_s = 8000 Hz then $N = f_s / f_a$ = 8 samples (assume *k* is the smallest positive integer, i.e. *k* = 1). Therefore the period length N = 8 samples.

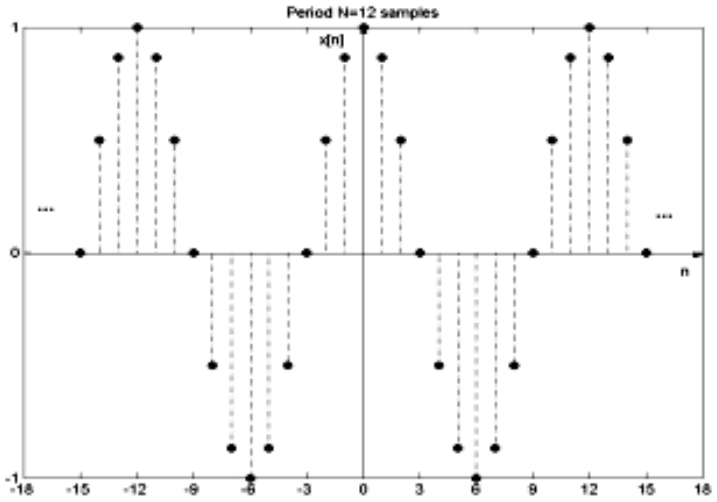
Note: A discrete - time sinusoid is periodic only when $\frac{f_a}{f_s}$ is a rational number. A rational number is a number that can be written as a ratio of two integers: $\frac{p}{q}$, p , q are integers and $q \neq 0$. For example: $\frac{1}{2}$, $\frac{3}{8}$, $\frac{2}{1}$ are all rational numbers.

Discrete-time signals: Periodic signals

Example 1:

- The following diagram depicts the cosine wave *with* period *N* = 12 samples.

$$x[n] = \cos\left(\frac{2\pi n}{12}\right)$$



Example 2:

- Determine the fundamental period of $x[n] = 10\cos\left(\frac{2\pi}{15}n + \frac{\pi}{3}\right)$

$$Digital\ frequency\ (\theta) = 2\pi/15$$

Recall the equation: $N = \frac{2\pi k}{\theta}$; k =1 is the smallest positive integer.

$$\therefore N = \frac{2\pi.1}{2\pi/15} = 15 \text{ samples}$$

Example 3:

- The sinusoidal signal *x[n]* has fundamental period *N=10* samples. Determine the smallest θ for which *x[n]* is periodic:

$$\theta = {}^{2\pi k}/_{N} = {}^{2\pi k}/_{10};$$

Smallest value of θ is obtained when *k* =1;

$$\therefore \theta = \frac{2\pi}{10} = \frac{\pi}{5} \text{ radians}$$

Example 4:

- Determine if or not the discrete-time signal, *x[n]*, below is periodic and if it is periodic, determine the fundamental period. $x[n] = \sin[\frac{2}{10}\,n + \pi]$.

Ans: $x[n] = \sin[0.2\,n + \pi] = \sin[0.2\,(n + N) + \pi]$ where N is an integer period.

∴ 0.2 N = 2 π k (k is an arbitrary integer and sine function is periodic with a period of 2 π.

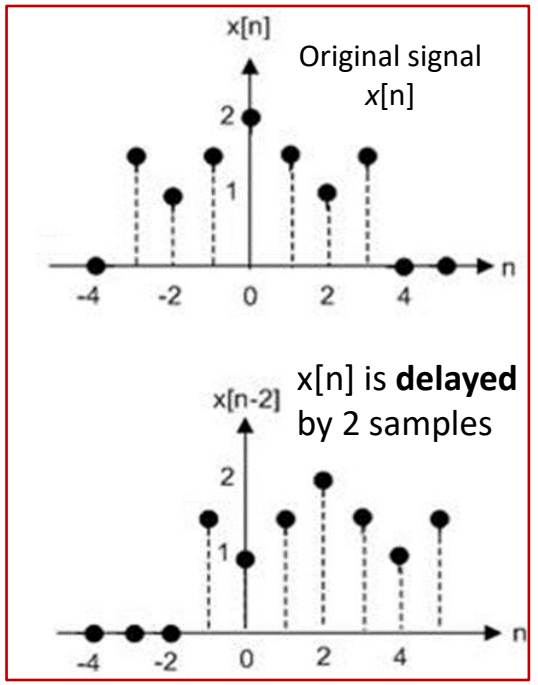
∴ N = 10 π k [non integer value of N exists]

Since π is an irrational number (cannot be expressed as a ratio of two integers) no positive integer value of N exists. Therefore, the signal is aperiodic.

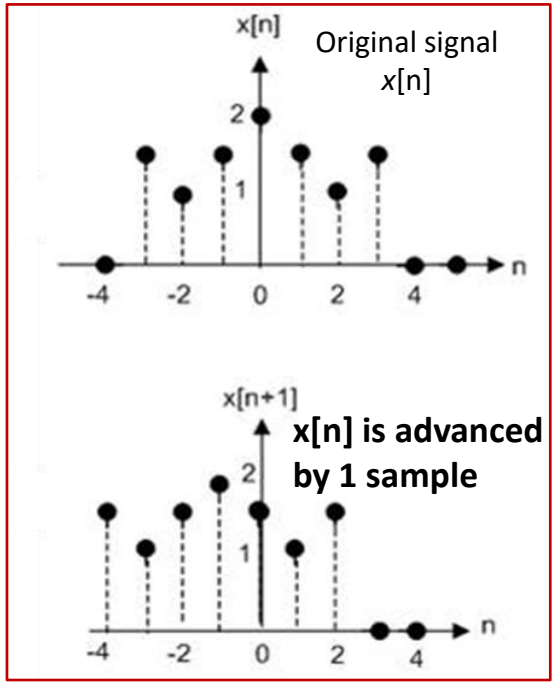
Discrete-time signals : Time Shifting

Time shifting of discrete-time signals

- A signal $x[n]$ may be shifted in time by replacing the independent variable n by n-k where k is an integer.
- If $k > 0 \rightarrow$ the time shift results in a **delay** of the signal by k samples {i.e. shifted to the right : $x[n-k]$}
- If $k < 0 \rightarrow$ the time shift results in an **advance** of the signal by k samples. {i.e. shifted to the left: $x[n+k]$}



Minus → Delay → Right

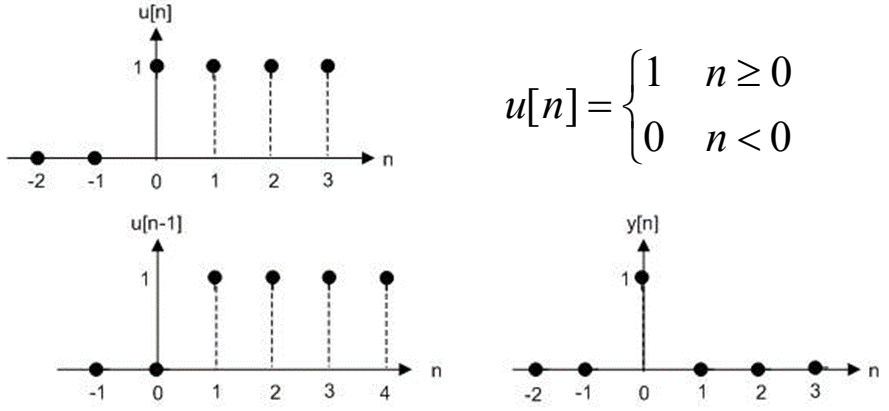


Plus → Advance → Left

Minus → Delay → Right

Example 5:

- Sketch the waveform : $y[n] = u[n] - u[n-1]$



$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Example 6:

- A continuous time signal is given by the following equation: $x(t) = 10e^{-t} - 5e^{-0.5t}$

  Write an equation for the discrete-time signal $x[n]$ and find $x[n-2]$.

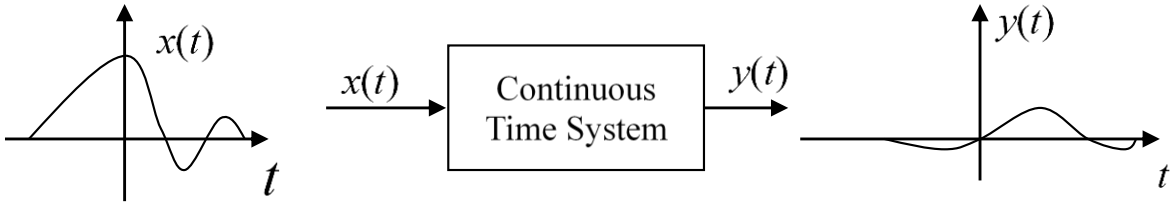
Ans: Substitute $t = nT$ in $x(t)$; [n - sample number & T – sampling period] $x[n] = 10\,e^{-nT} - 5e^{-0.5nT}$

$$x[n-2] = 10\,e^{-(n-2)T} - 5\,e^{-0.5(n-2)T}$$

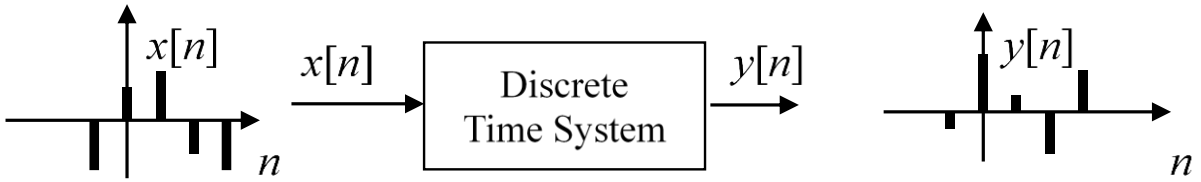
Difference equations

Systems

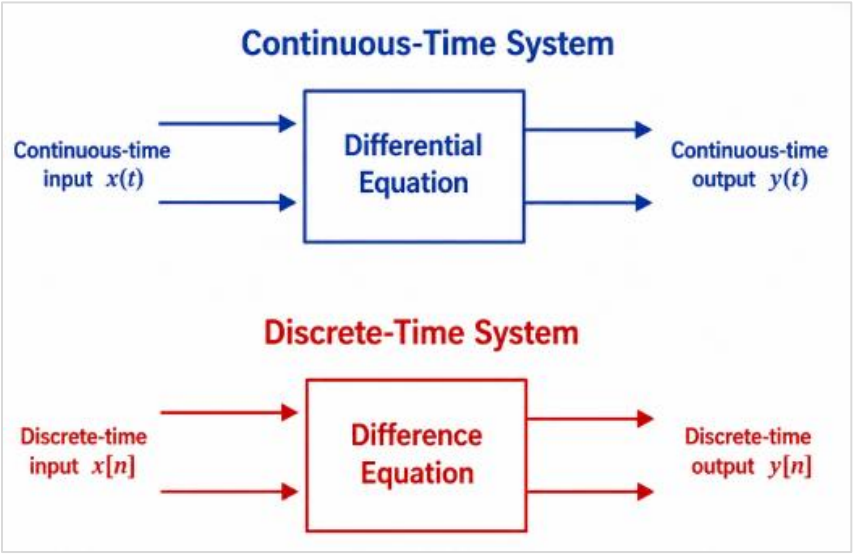
- A **continuous-time system** has a continuous-time input $x(t)$ and output $y(t)$, related by a mathematical rule:



- A **discrete-time system** has a discrete-time input $x[n]$ and output $y[n]$, related by a mathematical rule.



- Continuous-time systems are commonly described by **differential equations**, whereas discrete-time systems are described by **difference equations.**



Difference equations

Example 7:

- The RC circuit shown below is a continuous-time first-order low-pass system.
- If $e(t)$ is the input signal and $v_c(t)$ is the corresponding output signal, then we obtain using simple circuit analysis

$$\frac{dv_c(t)}{dt} + \frac{1}{RC}v_c(t) = \frac{1}{RC}e(t) \quad (1)$$

Differential equation

- Equation (1) can be approximated by a discrete-time system using the Backward Euler method.
- By substituting equation (2) into (1) and replacing t by nT, we obtain

$$\left.\frac{dv_c(t)}{dt}\right|_{t=nT} \approx \frac{v_c(nT) - v_c(nT-T)}{T} \quad (2)$$

- After substituting, the difference equation is:

$$\frac{v_C[n] - v_C[n-1]}{T} + \frac{1}{RC}v_C[n] = \frac{1}{RC}e[n]$$

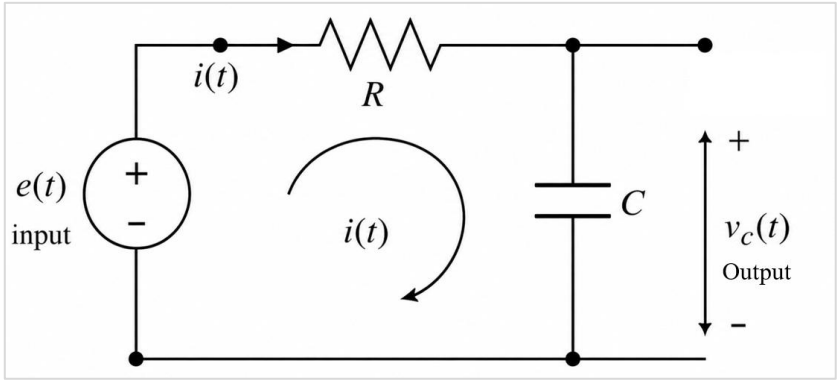
- Rearranging gives,

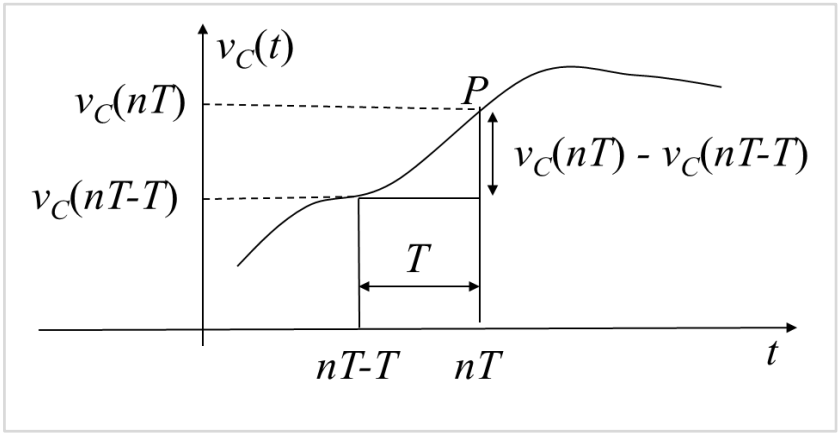
Difference equation

$$\underbrace{v_c[n]}_{\text{current output}} = \underbrace{\frac{RC}{RC+T}v_c[n-1]}_{\text{previous output}} + \underbrace{\frac{T}{RC+T}e[n]}_{\text{current input}}$$

The current output depends on the previous output and the current input.

RC circuit





Backward Euler approximation
[Assuming T is sufficiently small]

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Time and Frequency Domain Analysis: Fourier Transform

Continuous-time Fourier Transform (FT)

- The Fourier Transform (FT) represents a non-periodic continuous-time signal in the frequency domain.
- It allows us to analyse the frequency components present in a signal.
- The FT pair is given by:

Fourier Transform (FT): $X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$

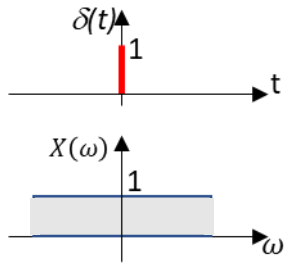
Inverse Fourier Transform (IFT): $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}d\omega$

- The FT converts the time domain signal, $x(t)$, into its frequency domain representation, $X(\omega)$, and the IFT converts the frequency domain representation, $X(\omega)$, back into the time domain $x(t)$.

Example 8:

- Find the Fourier Transform of *x(t) = δ(t).*

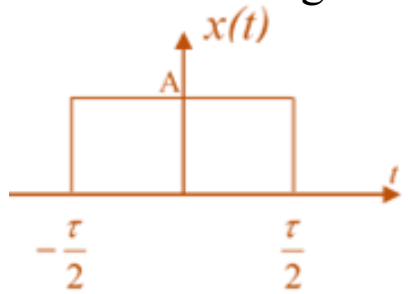
$$X(\omega) = \int_{-\infty}^{\infty} \delta(t)e^{-j\omega t}dt = 1$$



- The The unit impulse δ(t) has a constant Fourier Transform, X(ω)=1, for all frequencies.

Example 9:

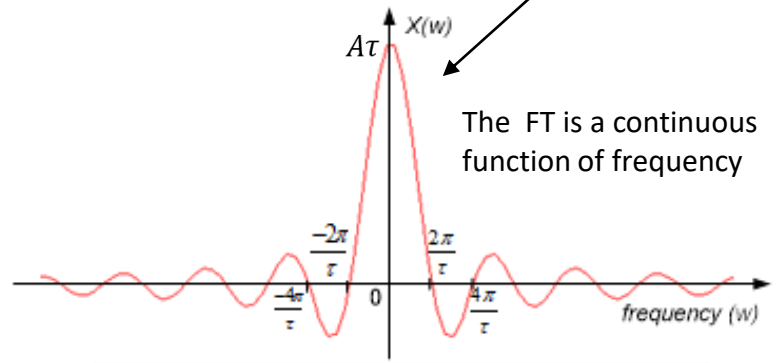
Evaluate the Fourier transform of a rectangular pulse shown below:



$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt$$

$$= \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} A\,e^{-j\omega t}\,dt \quad = -\frac{A}{j\omega}\left[e^{-\frac{j\omega\tau}{2}} - e^{\frac{j\omega\tau}{2}}\right]$$

$$X(\omega) = A\tau\frac{sin\frac{\omega\tau}{2}}{\frac{\omega\tau}{2}}$$ ← Sinc function



The FT is a continuous function of frequency

Narrower pulse in time → wider spectrum
Wider pulse in time → narrower spectrum

Fourier Transform Properties – Convolution in the Time Domain

For an LTI system, convolution describes how the output of a system depends on the **input signal and the** system's impulse response. If:

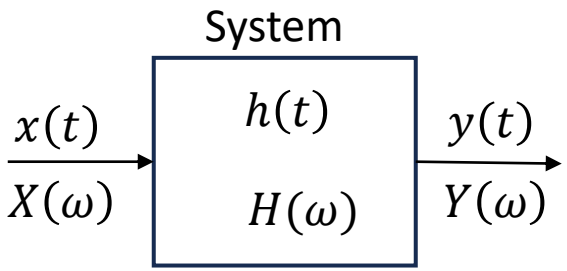
$x(t)$ = input signal

$h(t)$ = impulse response of a system

$y(t)$ = output signal

then the output is given by the convolution integral:

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$



$$y(t) = x(t) * h(t)$$

$$Y(\omega) = X(\omega).H(\omega)$$

Thus, the **convolution** in the **time domain** corresponds to **multiplication** in the **frequency domain**.

Convolution in the time domain

$$\int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau \overset{FT}{\longleftrightarrow} X(\omega).H(\omega)$$

Convolution in the Time Domain corresponds to multiplication in the frequency domain.

$$x(t) * h(t) \quad \leftrightarrow \quad X(\omega)\,.\,H(\omega)$$

$*$ *convolution*

Note:

When the input to an LTI system is a unit impulse ,

$$x(t) = \delta(t)$$

The output is the impulse response

$$y(t) = \delta(t) * h(t) = h(t)$$

Equivalently , in the frequency domain

$$X(\omega) = 1,$$

and therefore

$$Y(\omega) = \mathrm{X}(\omega).\ H(\omega) = H(\omega)$$

Taking IFT, we obtain $y(t) = h(t)$.

Time and frequency domain analysis

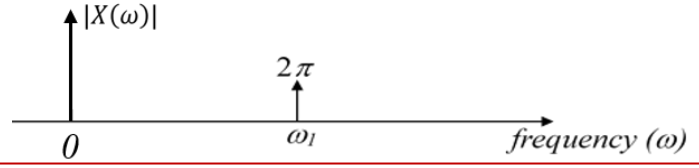
Example 10:

- Find the Fourier transform of $x(t) = e^{j\omega_1 t}$.

$$FT\{e^{j\omega_1 t}\} = \int_{-\infty}^{\infty} e^{j\omega_1 t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{j(\omega_1 - \omega)t} dt = 2\pi\delta(\omega_1 - \omega)$$

$$FT\{e^{j\omega_1 t}\} = X(\omega) = 2\pi\delta(\omega - \omega_1)$$

Hence the magnitude spectrum of $e^{j\omega_1 t}$ is shown below (impulse at $\omega = \omega_1$) :



Note: Use the following properties:

$\int_{-\infty}^{\infty} e^{j\omega t} dt = 2\pi\delta(\omega)$

and

$\delta(\omega - \omega_1) = \delta(\omega_1 - \omega)$

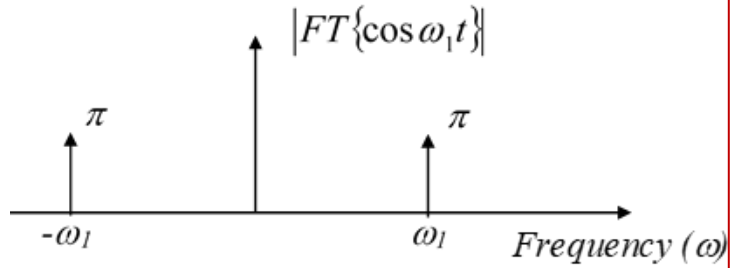
Example 11:

- Find the Fourier transform of $x(t) = \cos(\omega_1 t)$.

$$FT\{\cos\omega_1 t\} = FT\left\{\frac{e^{j\omega_1 t} + e^{-j\omega_1 t}}{2}\right\} = \frac{1}{2}FT\{e^{j\omega_1 t}\} + \frac{1}{2}FT\{e^{-j\omega_1 t}\}$$

$$= \pi\delta(\omega - \omega_1) + \pi\delta(\omega + \omega_1)$$ Use results from Example 10.

The magnitude spectrum of $\cos(\omega_1 t)$ is shown below:



- The spectrum of cosine signal has two impulses at ω=±ω1, each with weight π.

Fourier Transform Properties - Modulation

Modulation

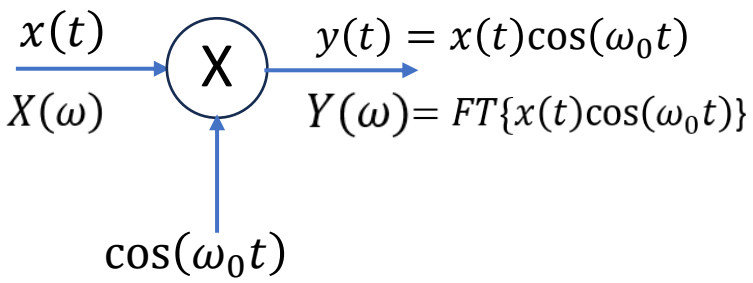
Modulation can be achieved by multiplying a signal $x(t)$ by a sinusoidal carrier $\cos(\omega_0 t)$. The modulated carrier becomes

$$y(t) = x(t)\cos(\omega_0 t)$$

The spectrum of the modulated signal consists of two shifted copies of $X(\omega)$, centred at $\pm\ \omega_0$, each scaled by 1/2:

$$FT\{x(t)\cos(\omega_0 t)\} = Y(\omega) = \frac{1}{2}X(\omega - \omega_0) + \frac{1}{2}X(\omega + \omega_0)$$

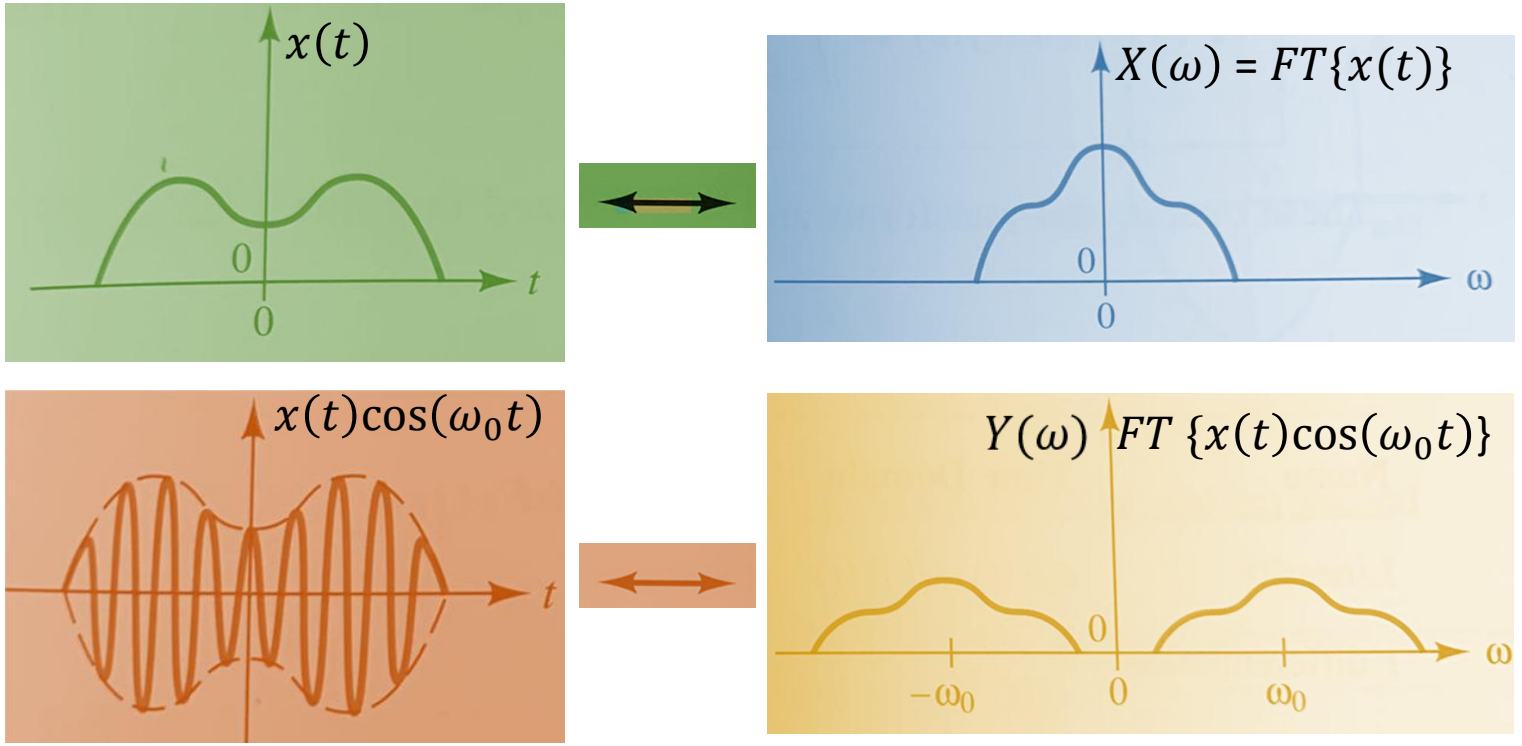
Modulation Scheme



When a signal $x(t)$ with spectrum $X(\omega)$ modulates a sinusoidal carrier having frequency ω_0, the resulting spectrum $Y(\omega)$ is the half the spectrum of $x(t)$ but shifted by $\pm\omega_0$

Multiplication in time	⇒	frequency shifting

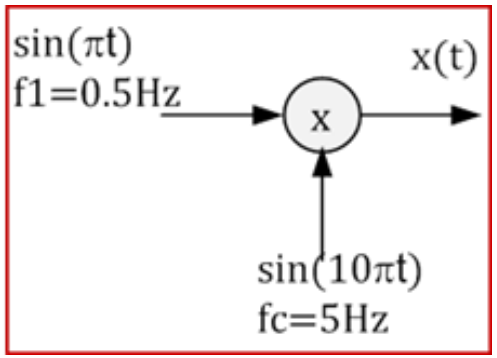
The effect of modulation is shown below:



Time and frequency domain analysis

Multiplication of sinusoids

- The product of two real sinusoids with frequencies f_1Hz and f_2Hz produces components at the sum and difference frequencies, (f_1+f_2)Hz and (f_1-f_2)Hz.



- Consider a signal $x(t)$ which is a product of two sinusoids, as given by:

$$x(t) = \sin(\pi t)\ .\ \sin(10\pi t)$$

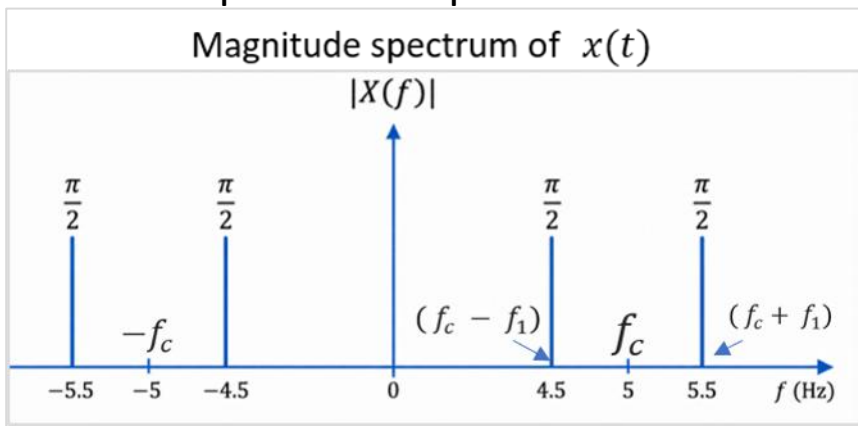
- $x(t)$ can be re-written as:

$$x(t) = \frac{1}{2}\cos(9\pi t) - \frac{1}{2}\cos(11\pi t) = \frac{1}{2}\cos(2\pi 4.5t) - \frac{1}{2}\cos(2\pi 5.5\,t)$$

- So the two frequencies are 4.5 Hz and 5.5 Hz. ($f_c + f_1$)
- Recall example 11: $FT\{\cos\omega_1 t\} = \pi\delta(\omega-\omega_1)+\pi\delta(\omega+\omega_1)$
- Using Example 11 and the expression for x(t), we can see that there are four spectral components located at :

$\pm(f_c - f_1) = \pm 4.5$ Hz and $\pm(f_c + f_1) = \pm 5.5$ Hz

Multiplication of two sinusoids → sum and difference frequencies



ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

- Discrete-time signals & difference equations (Ref pp. 9-19 and Ref pp. 43-59)
- Time and frequency domain analysis (Ref pp. 231-247 and Ref pp. 273-275)
- Sampling theorem & aliasing (Ref pp. 21-31 and Ref pp.396-413)
- A/D conversion & quantisation errors (Ref pp. 19-20 , Ref pp.31-37 and Ref pp.413-425)

Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

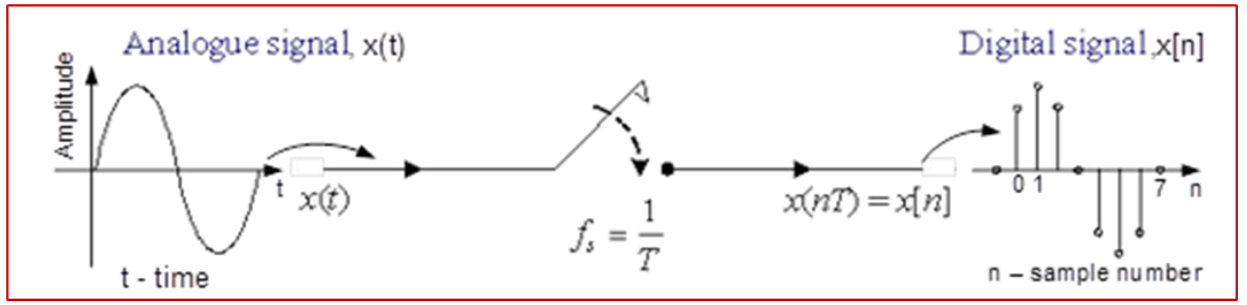
Sampling Theorem & Aliasing

Nyquist –Shannon Sampling theorem

- If the highest frequency in an analogue signal $x(t)$ is f_{max} , the signal must be sampled at a rate $f_s > 2\ f_{max}$ to avoid aliasing and allow the original signal to be reconstructed.
- The theoretical Nyquist condition is often written $f_s \geq 2\ f_{max}$, but sampling a sinusoid exactly at twice its frequency can be problematic depending on phase.
- The upper frequency limit of human hearing is generally taken to be approximately 20 kHz.
- In digital audio, a sampling rate of f_s = 44.1kHz is commonly used, which is greater than twice the approximate upper limit of human hearing (20 kHz).

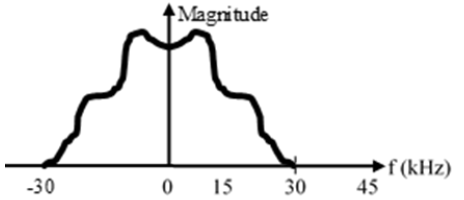
- For example, a sinewave of 50Hz (shown on the right) is sampled at a frequency 100 Hz (Nyquist rate) (i.e. 100 samples per second) . Can the original signal be reconstructed from these samples?

Acquiring discrete-time signal, x[n] by sampling continuous-time signal x(t).

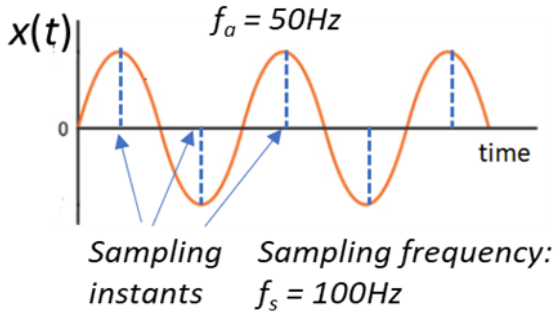


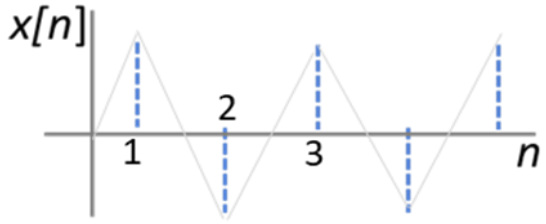
Example 12:

- What is the Nyquist rate for this signal??



a) 30 kHz
b) 60 kHz
c) 45 kHz
d) None of the above





Aliasing: Aliasing is a distortion of the sampled signal caused by an insufficient sampling rate, making it impossible to correctly reconstruct the original signal.

Sampling Theorem & Aliasing

Overview of a DSP system

- An analogue signal processing system is shown in Figure 5, in which both the input signal and output signal are in analogue form.
- A digital signal processing system shown in Figure 6 provides an alternative method for processing the analogue signal.

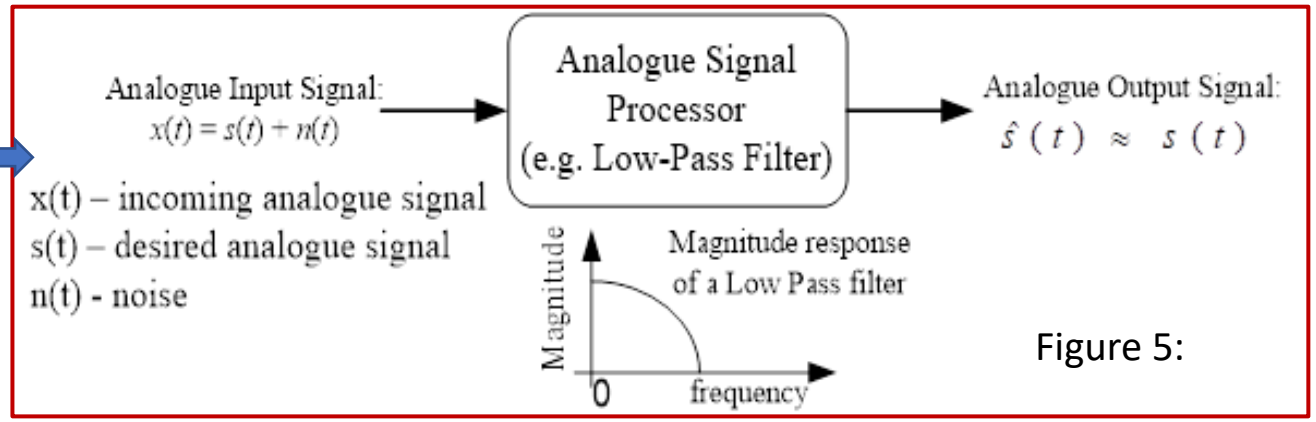


Figure 5:

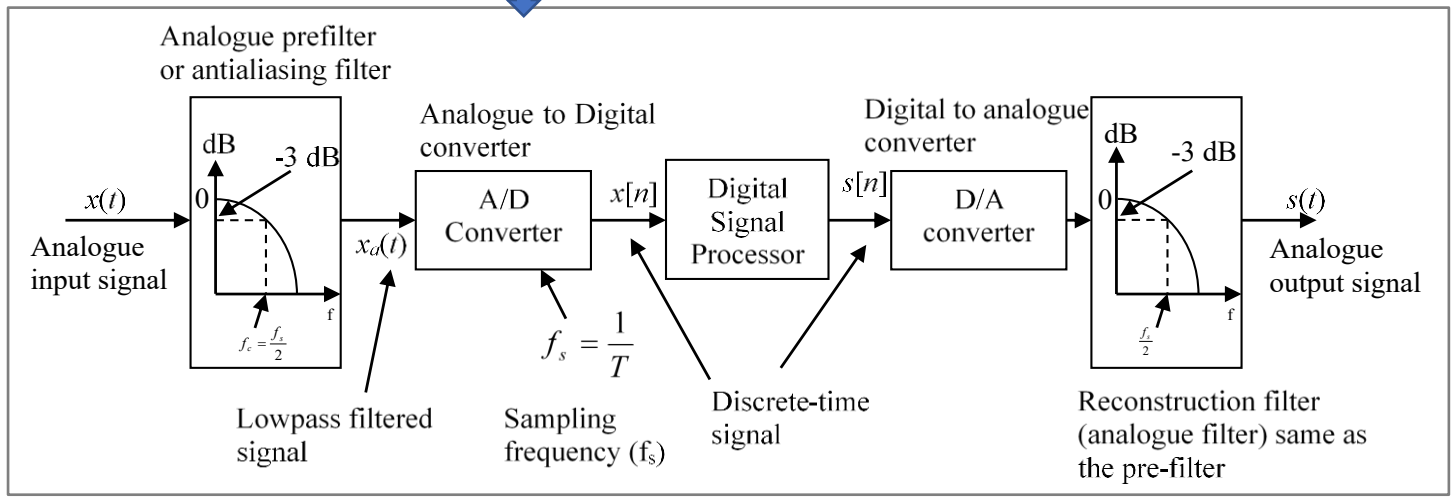


Figure 6: A general process of converting analogue signals into digital signals and back to analogue form.

The Digital Signal Processor performs the required operation on the sampled signal x[n], such as filtering, enhancement, detection or other DSP algorithms, to produce the output s[n].

Analogue signal → Anti-aliasing filter → A/D → DSP → D/A → Reconstruction filter → Analogue output

Name of the Block	Function
Anti-aliasing filter ($f_c > \frac{f_s}{2}$)	Band-limits the analogue input signal before sampling so that frequency components above the Nyquist frequency fs/2 do not cause aliasing.
Analogue-to-digital converter (A/D)	Converts the analogue input signal into a digital signal by sampling and quantisation.
Digital Signal Processor (heart of the system)	To process the digital signal according to the pre-defined rules.
Digital-to-analogue converter (D/A)	Converts the digital signal into an analogue output signal.
Reconstruction filter ($f_c > \frac{f_s}{2}$)	Smooths the output of the D/A converter and removes unwanted high-frequency components.

Table 1: Description of blocks contained in a general DSP process

Sampling Theorem & Aliasing

Sampling of Continuous –time signal

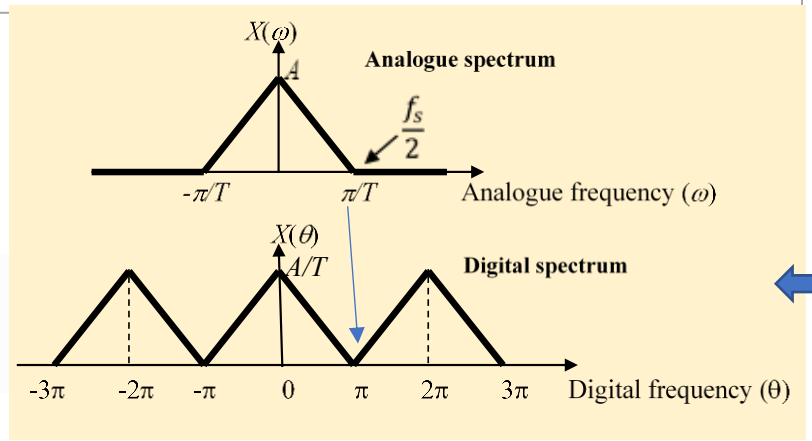
- If a continuous-time signal $x(t)$ is sampled every T seconds, then at the output of the Analogue to Digital Converter, as shown below we obtain a discrete-time signal x[n]



- where t = time; n = sample number; ω = analogue frequency; θ = digital frequency ($-\pi \leq \theta \leq \pi$);
- $\theta = \omega T$, $\theta = 2\pi \frac{f_a}{f_s}$, $X(\theta)$ = digital spectrum; $X(\omega)$ = analogue spectrum.

Note:
$\omega = \frac{\pi}{T}$ (or $f = \frac{f_s}{2}$) corresponds to $\theta = \pi$

$$f = \frac{f_s}{2} \iff \theta = \pi$$



Discrete-Time Fourier Transform Pair

- The Discrete-Time Fourier Transform (DTFT) pair is given by:

Discrete Time Fourier Transform (DTFT):
$$X(\theta) = \sum_{n=-\infty}^{\infty} x[n] e^{-jn\theta}$$

Inverse Discrete Time Fourier Transform (IDTFT):
$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\theta) e^{jn\theta} d\theta$$

$$where \ -\pi \leq \theta \leq \pi, \quad \theta = \omega T, \quad \theta = 2\pi \frac{f_a}{f_s}$$

- Using Inverse Fourier Transforms in Continuous-time and discrete-time domains, we can show that

$$\boldsymbol{X(\theta) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T})}$$

- It is seen that the DTFT $X(\theta)$ is periodic with period 2π. Sampling produces repeated copies of the original analogue spectrum in the digital-frequency domain. **If these spectral copies overlap, aliasing occurs.**

Sampling Theorem & Aliasing

Aliasing

- Aliasing arises when a continuous-time signal is sampled at a rate that is insufficient to capture the changes in the signal. If aliasing occurs, the original continuous time signal cannot be recovered.
- The sampling frequency f_s should be greater than twice the highest frequency f_{max} in the signal to avoid aliasing.
- Figure 7 illustrates the relationship between the digital spectrum $X(\theta)$ and the analogue spectrum $X(\omega)$ for the case $X(\omega) = 0$ where $|\omega| \geq \frac{\pi}{T}$ (i.e. $|f| \geq \frac{f_s}{2}$)

Case 1: $X(\omega) = 0$, $|\omega| \geq \frac{\pi}{T}$; (Sampling theorem holds)

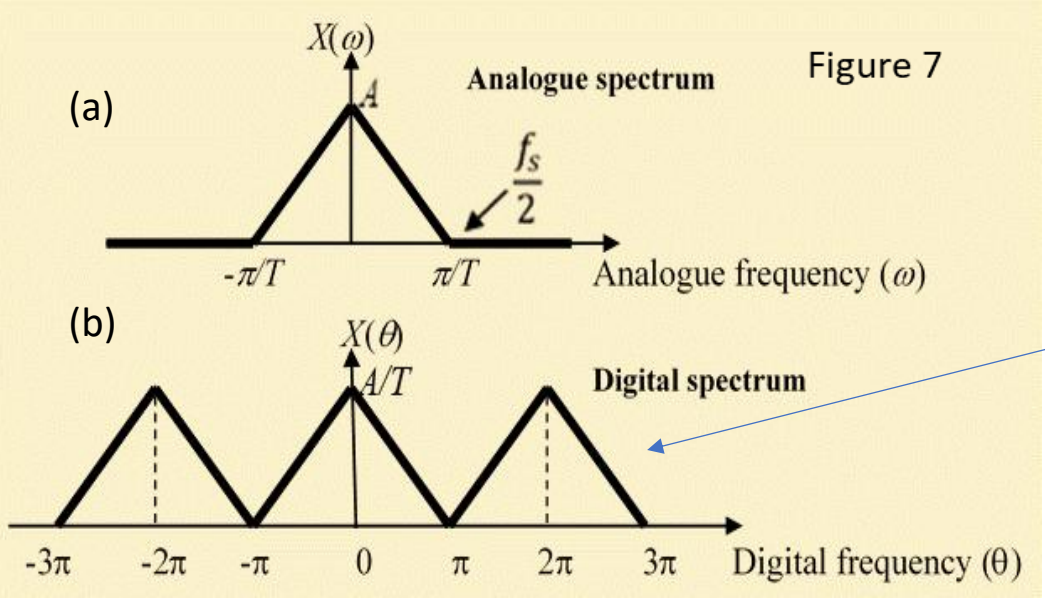


Figure 7

$$X(\theta) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T})$$

Sampling produces repeated copies of the original analogue spectrum. When these spectral replicas do not overlap, there is no aliasing.

Case 1: No aliasing $f_{max} < \frac{f_s}{2}$

Spectral copies do not overlap → original signal can be reconstructed.

Sampling Theorem & Aliasing

Case 2: Aliasing

- $X(\omega) = 0$, $|\omega| \geq \frac{3\pi}{2T}$; This means the analogue spectrum extends to $\omega_{max} = \frac{3\pi}{2T}$
- Since $\omega = \frac{2\pi}{T}$ or $(f = f_s)$, the Nyquist frequency is $\omega = \frac{\pi}{T}$ or $(f = \frac{f_s}{2})$
- Therefore $\frac{3\pi}{2T} > \frac{\pi}{T}$; so the signal exceeds the Nyquist frequency

Case 2: $X(\omega) = 0$, $|\omega| \geq \frac{3\pi}{2T}$; (Sampling theorem does not hold)

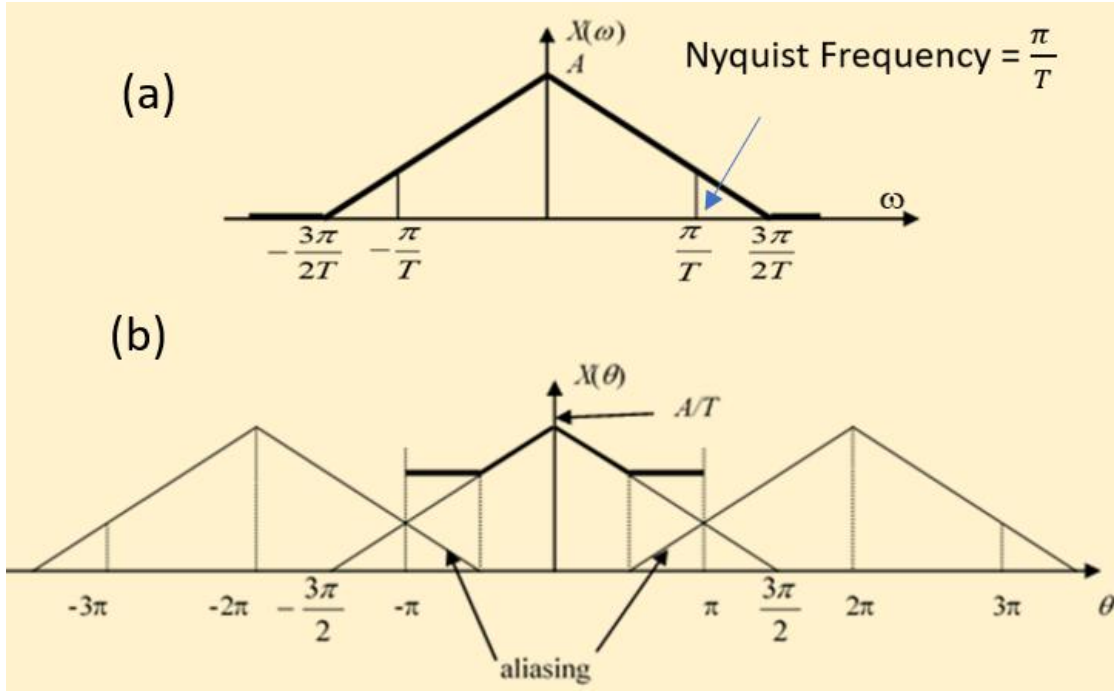


Figure 8: (a) Analogue spectrum extending beyond the Nyquist frequency. (b) After sampling, adjacent spectral replicas overlap, resulting in aliasing.

Case 2: Aliasing $f_{max} > \frac{f_s}{2}$

Spectral copies overlap → original signal cannot be correctly reconstructed.

$$X(\theta) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(\omega + k\frac{2\pi}{T})$$

Sampling Theorem & Aliasing

Aliasing

- The digital spectrum consists of original band limited analogue spectrum and repeats at multiples of the sampling frequency f_s (refer to Figure 9) as given by:

$$f_k = f_0 + k f_s$$

- where k is an integer (k = ±1, ±2, ±3, …) and f_0 is the frequency component present in the fundamental region ($\frac{-f_s}{2} \leq f \leq \frac{f_s}{2}$) of the original analogue spectrum.
- For $k \neq 0$, the repeated spectral components lie outside the fundamental frequency range but correspond to the same digital frequency.

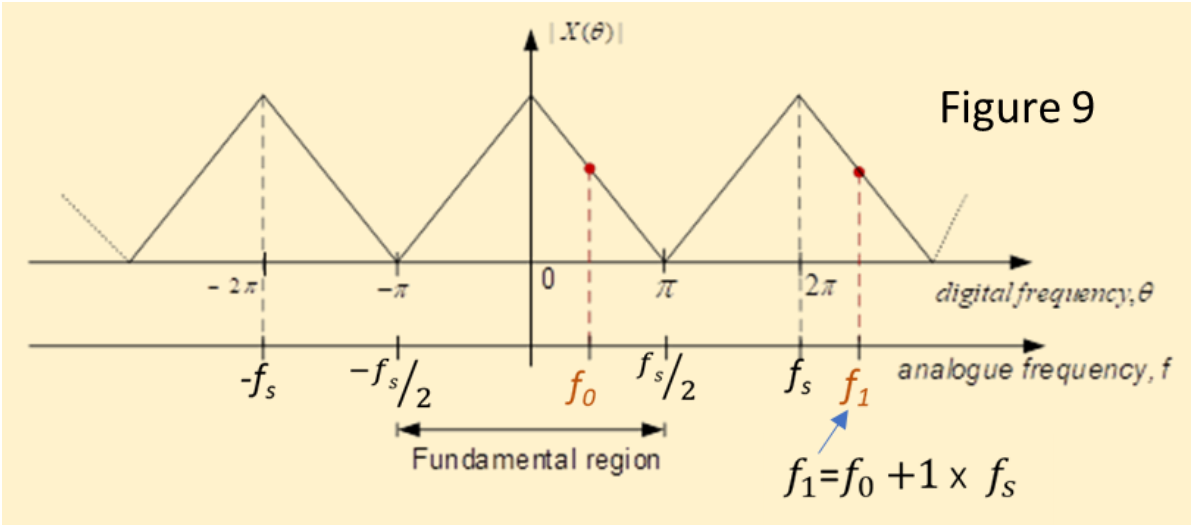


Figure 9

Example 13:

- Consider the following analogue signal:

$x(t)$ = 3 cos(50π t) + 10 sin (300π t) – cos (100π t) ;

What is the Nyquist rate required to sample this signal?

Ans: The frequencies present in the signal above are:

f_1 = 25Hz; f_2 = 150 Hz; f_3 = 50 Hz;

Hence, f_{max} = 150 Hz

The Nyquist rate is $f_N = 2 f_{max} = 300$ Hz.

Note: In practice, sampling frequency f_s chosen greater than the Nyquist rate.

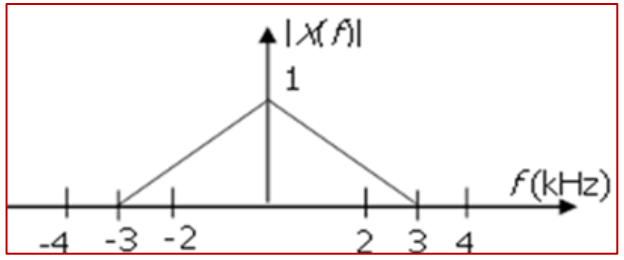
Sampling Theorem & Aliasing

Example 14:

The magnitude spectrum of a continuous-time signal $x(t)$ is shown below. The signal is bandlimited to 3kHz. If the signal is sampled at $f_s = 2kHz$,

(i) Sketch the resulting digital spectrum $X(\theta)$

(ii) Does aliasing occur? Explain your answer.



Equation for the triangular spectrum :

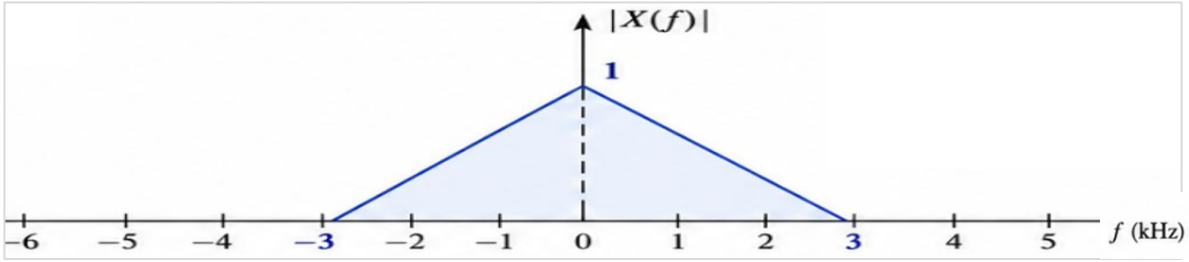
$$|X(f)| = \begin{cases} 1 - \dfrac{|f|}{3}, & |f| \le 3 \text{ kHz} \\ 0, & |f| > 3 \text{ kHz}. \end{cases}$$

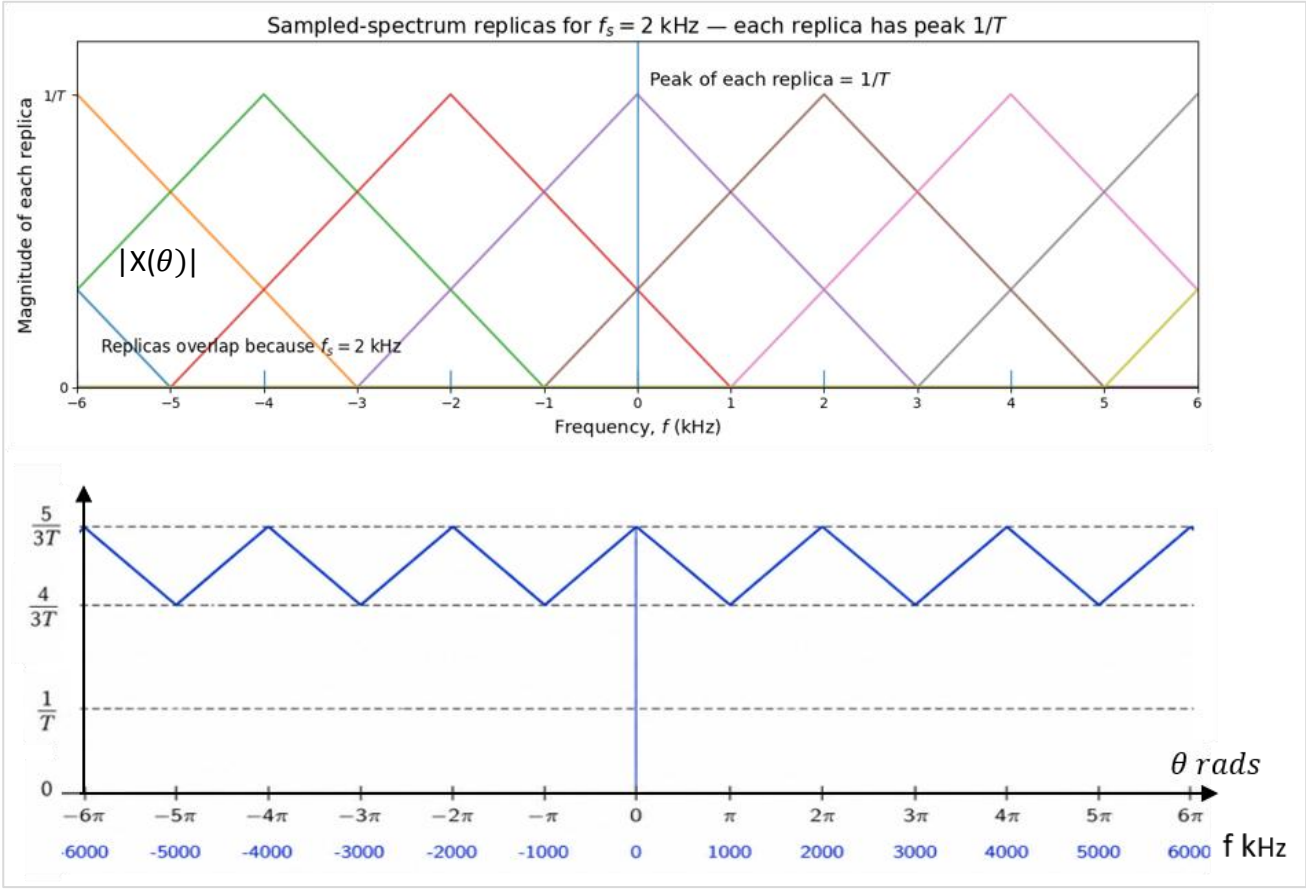
Equation for the triangular spectrum shifted from 0 to 2kHz:

$$|X(f)| = \begin{cases} 1 - \dfrac{|f-2|}{3}, & -1 \le f \le 5 \text{ kHz} \\ 0, & \text{otherwise}. \end{cases}$$

Equation for the triangular spectrum shifted from 0 to -2kHz:

$$|X(f)| = \begin{cases} 1 - \dfrac{|f+2|}{3}, & -5 \le f \le 1 \\ 0, & \text{otherwise}. \end{cases}$$





Because the spectral replicas overlap, the original signal cannot be recovered exactly from the samples.

Sampling Theorem & Aliasing

Example 15:

- Consider the following analogue signal

$$x(t) = 3\,cos(2000\pi t) + 5\,sin(6000\pi t) + 10\,cos(12000\pi t)$$

(a) What is the Nyquist rate for this signal?

The frequencies present in the analogue signal are:

f_1 = 1 kHz; f_2 = 3 kHz; f_3 = 6 kHz

Thus f_{max} = 6 kHz .

According to the sampling theorem, $f_s > 2\,f_{max}$ = 12 kHz

The Nyquist rate is = 12 kHz.

(b) Now assume that we sample this signal $x(t)$ using a sampling rate f_s = 5 kHz (samples/sec). What is the discrete-time signal obtained after sampling?

$$f_s = 5kHz \Rightarrow \frac{f_s}{2} = 2.5kHz$$

Recall slide 27: $\boldsymbol{f_k = f_0 + kf_s}$ [f_0 is the frequency present in the fundamental region of the original analogue spectrum] . Using this equation, we can calculate the new frequency ($\boldsymbol{f'_n}$) caused by aliasing effect as follows:

$$\boldsymbol{f'_n = f_n - kf_s} \qquad -\frac{f_s}{2} \le \boldsymbol{f'_n} \le \frac{f_s}{2}; \qquad (k = \pm1, \pm2, \ldots)$$

$\boldsymbol{f'_n}$ - new frequency due to aliasing ; $\boldsymbol{f_n}$ – frequency causing the aliasing (which is above the folding frequency)

- The frequency f_1 = 1000 Hz is $< \frac{f_s}{2}$ (= 2500 Hz) and thus it is not affected by aliasing.
- However, the other two frequencies f_2 & f_3 are above the folding frequency and will be affected by aliasing .

$f_2' = f_2 - 1\,f_s = 3000 - 5000 = -2$ kHz

$f_3' = f_3 - 1\,f_s = 6000 - 5000 = 1$ kHz

Therefore,

$$x[n] = 3\,cos\left(2\pi\left(\frac{1000}{5000}\right)n\right) + 5\,sin\left(2\pi\left(-\frac{2000}{5000}\right)n\right) + 10\,cos\left(2\pi\left(\frac{1000}{5000}\right)n\right)$$

(C) What analogue signal y(t) can we reconstruct from the samples using ideal interpolation?

After aliasing, the sampled signal contains frequency components at 1 kHz and 2 kHz. Therefore, the analogue signal we can reconstruct is

$$y(t) = 13\cos(2000\pi t) - 5\sin(4000\pi t)$$

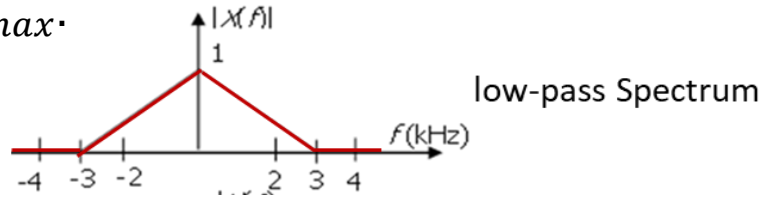
which is different from the original signal $x(t)$.

The reconstructed signal $y(t)$ is different from the original signal because the sampling rate is too low, causing aliasing.

Sampling Theorem & Aliasing – Bandpass Spectrum

Sampling of Bandpass Signal

- So far, we have considered **low-pass signals**, where the spectrum extends from DC (0 Hz) to a maximum frequency f_{max}.

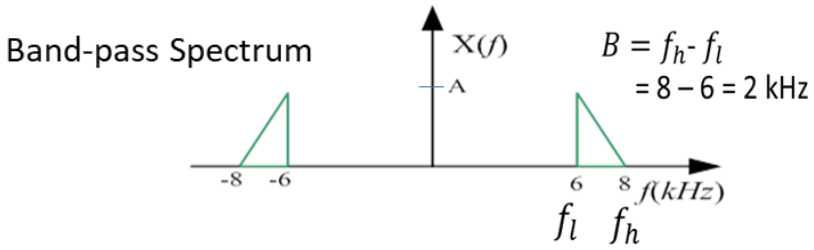


low-pass Spectrum

- For a low-pass signal, the sampling theorem requires

$$f_s > 2f_{max}$$

- A **bandpass signal** is different. Its spectrum occupies only a band of frequencies from f_l to f_h, and does not contain frequencies near DC.



- Bandwidth B = $f_h - f_l$ = 8−6 = 2 kHz.
- For a bandpass signal, the sampling rate can sometimes be based on its bandwidth B, rather than its highest frequency f_h.

- For the above Band-pass spectrum , ordinary Nyquist sampling would require

$$f_s > 2f_h = 16\ kHz.$$

- However, with bandpass sampling , it may be possible to use a much lower sampling rate.
- The sampling frequency (f_s) must be chosen carefully so that the spectral replicas do not overlap.
- A particularly convenient special case of the **bandpass sampling theorem** occurs when the minimum sampling frequency is $f_s = 2\ B$

  where B = $f_h - f_l$ is the bandwidth
- For the above signal B = 8−6=2 kHz, so

$$f_s = 2\ B\ = 4kHz$$

  which is much lower than the 16 kHz required for ordinary Nyquist sampling.
- The minimum value $f_s = 2\ B$ can be used without aliasing when f_h is an integer multiple of the bandwidth B.

Sampling Theorem & Aliasing – Bandpass Spectrum

Sampling of Bandpass Signal

- $f_s = 4kHz$; The new frequency due to aliasing is:

$$f_o' = \mathrm{f_o} - \mathrm{kf_s}, \qquad \mathrm{k} = 0, \pm1, \pm2, \ldots$$

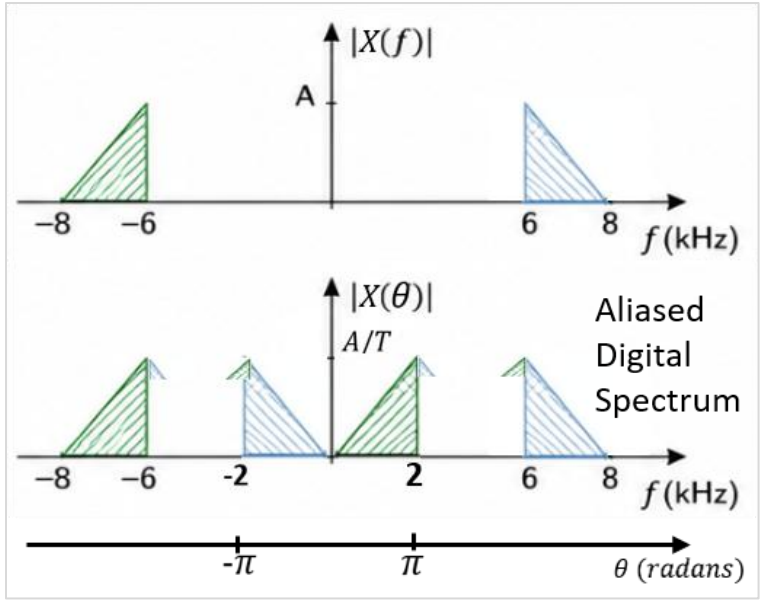
f_0 is the frequency present in the original analogue spectrum

For k = 2: $f_o' = 6 - 2(4) = -2;$

$f_o' = 8 - 2(4) = 0$

For k = −2: $f_o' = -6 + 2(4) = 2;$

$f_o' = -8 + 2(4) = 0$



Example 16:

- Consider a bandpass signal, *s(t)*, whose magnitude spectrum is

$$|\mathrm{S(f)}| = \begin{cases} 3 - |\mathrm{f}| & 2\mathrm{kHz} < |\mathrm{f}| < 3\mathrm{kHz} \\ 0 & \text{otherwise} \end{cases}$$

where *f is* in kHz. The signal *s(t)* is sampled uniformly with a sampling frequency of 2kHz. Sketch the digital spectrum $|\mathrm{S}(\theta)|$.

The new frequency due to aliasing is: $f_o' = \mathrm{f_o} - \mathrm{kf_s},\ \mathrm{k} = 0, \pm1, \pm2, \ldots$

f_0 is the frequency present in the original analogue spectrum.

For k = 1:

$f_o' = 2 - 1(2) = 0$

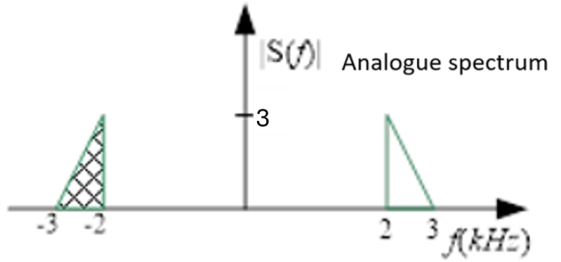
$f_o' = 3 - 1(2) = 1$

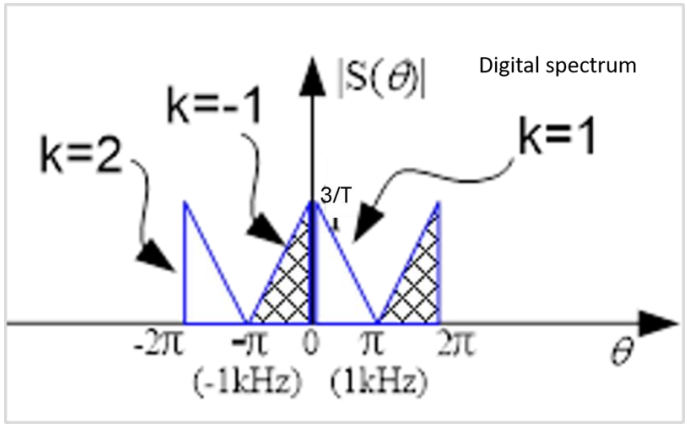
For k = −1:

$f_o' = -2 + 1(2) = 0$

$f_o' = -3 + 1(2) = -1$

Choosing $f_s = 2\mathrm{kHz} = 2\mathrm{B}$ (with B = 1kHz) allows bandpass sampling without overlap of replicas.





ELEC3104: Digital Signal Processing

Topic 1: DSP fundamentals

- Discrete-time signals & difference equations (Ref pp. 9-19 and Ref pp. 43-59)
- Time and frequency domain analysis (Ref pp. 231-247 and Ref pp. 273-275)
- Sampling theorem & aliasing (Ref pp. 21-31 and Ref pp.396-413)
- A/D conversion & quantisation errors (Ref pp. 19-20 , Ref pp.31-37 and Ref pp.413-425)

Reference Book: J. Proakis & D. Manolakis, Digital Signal Processing, Prentice-Hall, 2014

Prof. Eliathamby Ambikairajah, School of EE&T

Term 3, 2026

A/D Conversion and Quantisation

Analogue to Digital (A/D) Conversion Process

- Before a signal can be processed using DSP algorithms, the analogue signal must first be converted into digital form. [$x_a(t)$ to $x[n]$ transformation – see Figure 10.].
- The A/D conversion process involves three main steps:
  1. **Sampling:** The band-limited analogue signal is sampled, at regular time intervals T, producing a discrete-time signal
  2. **Quantisation:** The amplitude of each sample is quantised to one of 2^B discrete levels (where *B* is the number of bits per sample)
  3. **Encoding:** Each quantised level is represented by a unique B-bit binary word.

A practical representation of the A/D conversion process is shown in Figure 11.

- The sample-and-hold (S/H) circuit takes a sample of the analogue signal every T seconds and holds that value constant until the next sample is taken.

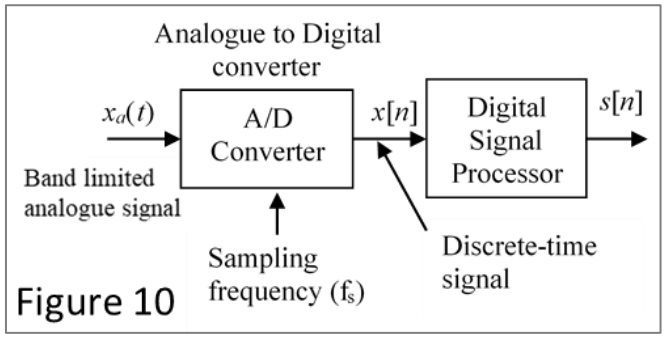


Figure 10

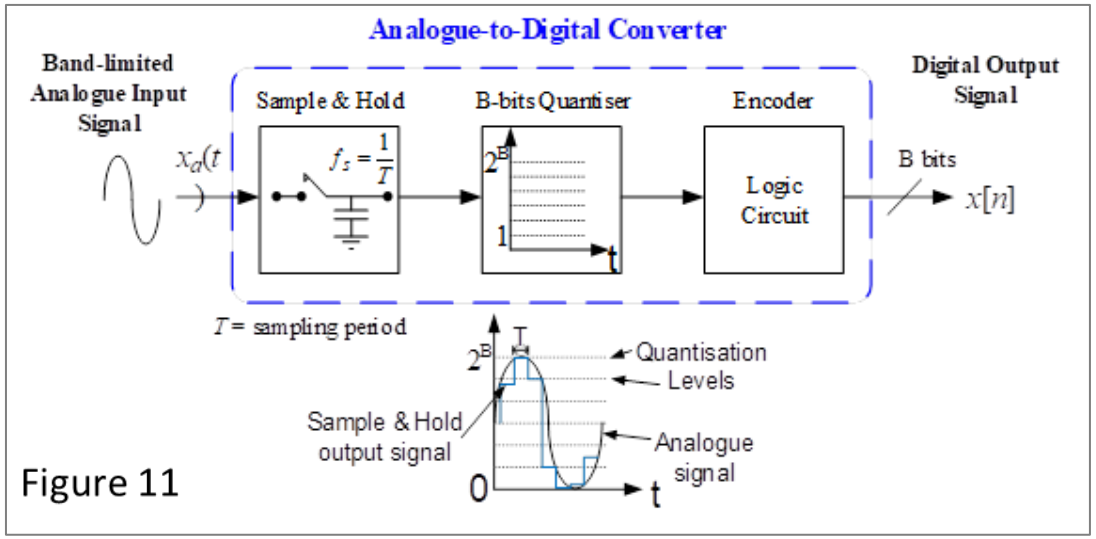


Figure 11

Quantisation errors

Quantisation and Encoding

- **Quantisation** maps each sampled analogue amplitude to one of a finite number of discrete amplitude levels.
- For a B-bit A/D converter, there are $L = 2^B$ possible quantisation levels.

For example, a 3-bit A/D converter has $2^3 = 8$ levels, as shown in Figure 12.

- Each quantised level is then encoded as a B-bit binary number.
- Since the actual sample amplitude is generally replaced by the nearest available quantisation level, **quantisation introduces an error**. This error cannot be removed after quantisation.
- A/D converters are commonly described by their resolution, or number of bits (e.g. 8-bit, 12-bit or 16-bit). More bits provide more quantisation levels and therefore finer amplitude resolution.
- Figure 13 shows the input-output characteristic of a **4-bit bipolar A/D converter** using linear quantisation and two's-complement coding.

An 8-level unipolar A/D converter coding scheme

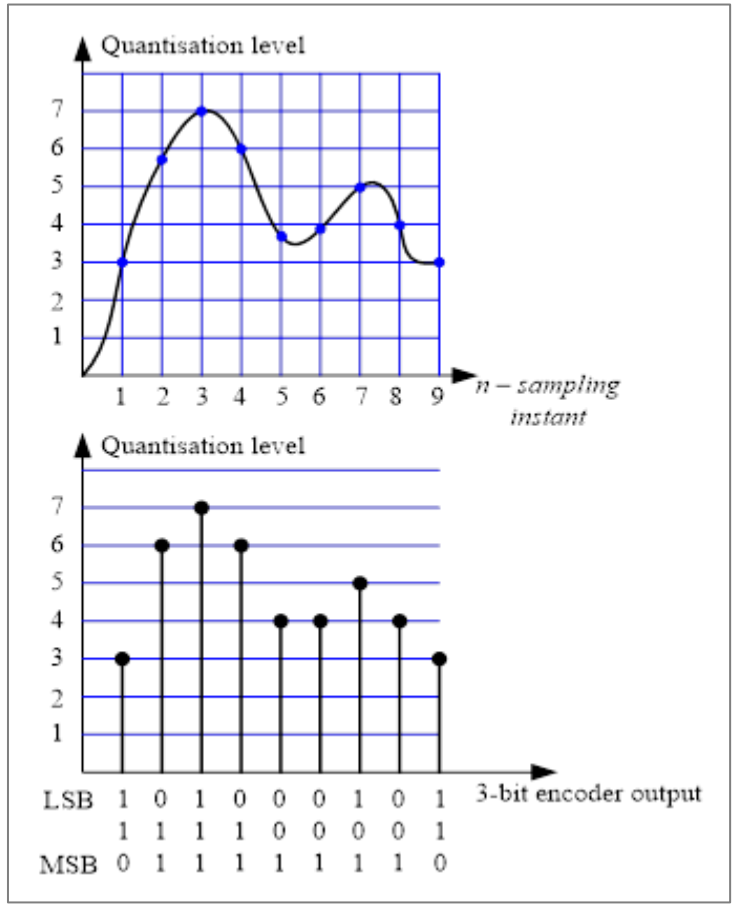


Figure 12: Quantisation of discrete-time signals

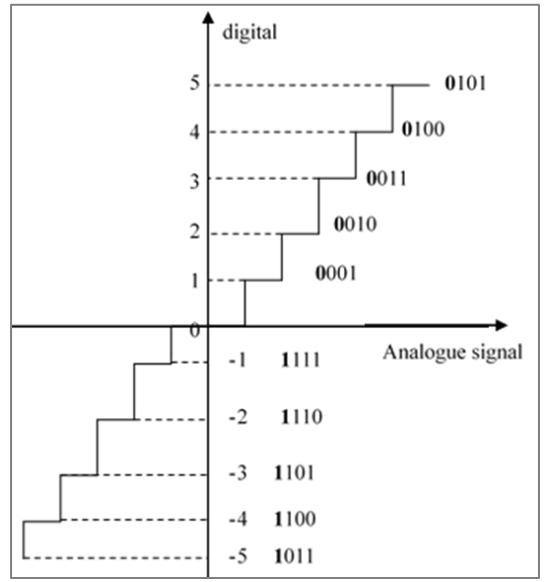


Figure 13: A 4 bit bipolar A/D converter with 2's complement coding scheme

Quantisation: continuous amplitude → discrete amplitude level
Encoding: discrete level → binary word

Quantisation errors

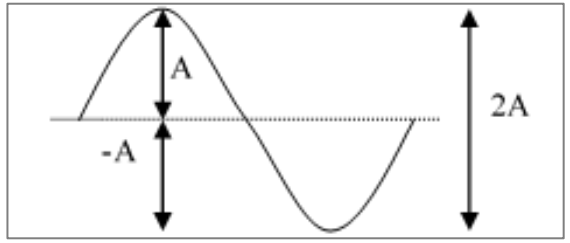
Quantisation step size

- For a B-bit A/D converter, the number of quantisation levels (L) is 2^B. The interval between adjacent levels, known as the quantisation step size or resolution (ΔV), is given by:

  $\Delta V = \frac{V}{2^{B-1}} \approx \frac{V}{2^B}$ [The approximation holds when B is Large (Normally B >10 bits)

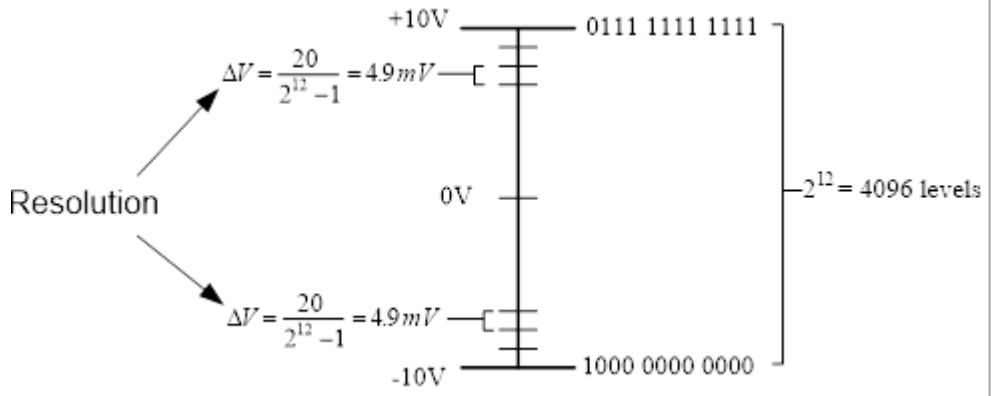
- where ΔV = resolution, B = number of bits and V = full scale (peak-to-peak) input voltage range.

- For a sine wave input of amplitude A, the peak-to-peak amplitude is 2A. Therefore ,the quantisation step size becomes: $\Delta V \approx \frac{2A}{2^B}$



Example 17:

- A 12 bit A/D converter (bipolar) with an input voltage range of ±10V will have a least significant bit (LSB) of $\frac{20V}{2^{12}} mV = 4.88mV$ (resolution or quantisation step size, ΔV)



Quantisation errors

Quantisation error (e)

- Quantisation introduces an error during the A/D conversion process.
- The quantisation error, $e[n]$, is the difference between the actual amplitude of the input signal and the quantised amplitude:

$$e[n] = x[n] - x_q[n]$$

where $x[n]$ is the actual amplitude and $x_q[n]$ is the quantised amplitude

- Figure 14 shows a signal and its quantised representation. The difference between them is the quantisation error e[n], which appears as quantisation noise (green plot) .
- The quantisation error appears as noise in the reconstructed analogue signal and is therefore also called quantisation noise.
- Quantisation error can be reduced by increasing the number of bits B. More bits produce smaller quantisation steps and therefore smaller quantisation errors.
- For an A/D converter with bipolar signal inputs, the quantisation error (e[n]) is bounded between $-\Delta V/2$ and $+\Delta V/2$ (see Figure 15). Quantisation error is commonly modelled as random noise uniformly distributed between −ΔV/2 and +ΔV/2, with zero mean.
- We can't use an infinite number of bits for A/D conversion, so some quantisation noise will always be present. For human perception (e.g. for audio signals), when **16 bits or more** are used for each sample, the associated quantisation noise generally becomes imperceptible

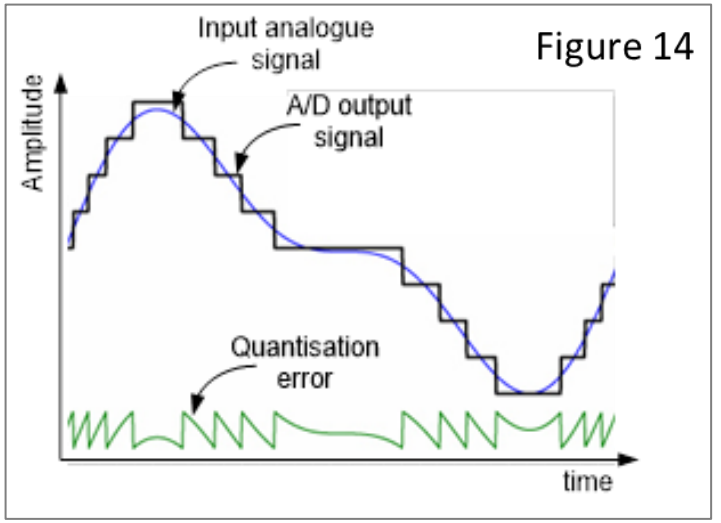


Figure 14

Quantisation (error) noise is random and uniformly distributed with zero mean.

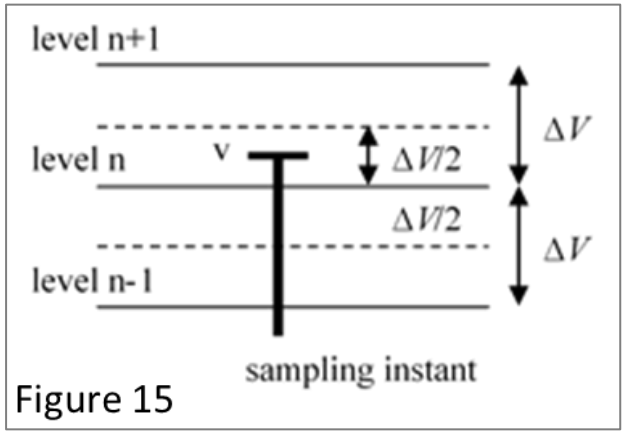


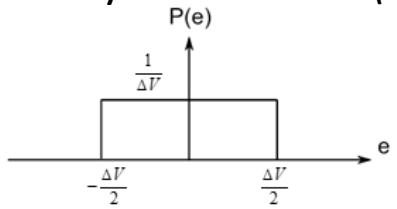
Figure 15

Maximum Quantisation error (e[n]) = $\frac{\Delta V}{2}$

Quantisation errors

Quantisation error (e)

- Assuming the quantisation error is uniformly distributed between $-\Delta V/2$ and $+\Delta V/2$, its probability density function is $P(e)=1/\Delta V$.



- The quantisation noise power or variance $\sigma_e{}^2$ is hence given by:

$$\sigma_e^2 = \int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 P(e)de$$

$$= \frac{1}{\Delta V}\int_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} e^2 de = \frac{1}{\Delta V}\left[\frac{e^3}{3}\right]_{-\frac{\Delta V}{2}}^{\frac{\Delta V}{2}} = \frac{\Delta V^2}{12}$$

Note: Probability of quantisation error is constant:

$P(e)=1/\Delta V$

- For uniform quantisation [all steps (ΔV) are of equal size], the quantisation noise power:

$\sigma_e^2 = \frac{\Delta V^2}{12}$ {**rms** quantisation error $\boldsymbol{\sigma_e}= \Delta V/\sqrt{12}$ }

Signal-to-Quantisation Noise ratio (SQNR)

- Signal-to-quantisation noise power ratio (SQNR) is defined as:

$$SQNR = \frac{P_s\ (signal\ power)}{P_n\ (noise\ power)}$$

$$P_s = \frac{1}{N}\sum_{n=1}^{N}\{x[n]\}^2; \quad P_n = \frac{1}{N}\sum_{n=1}^{N}\{e[n]\}^2; \quad P_n = \sigma_e^2 = \frac{\Delta V^2}{12}$$

$$\therefore SQNR(dB) = 10\,log\frac{P_s}{P_n} = 10\,log\frac{\sum_{n=1}^{N} x^2[n]}{\sum_{n=1}^{N} e^2[n]}$$

- The dynamic range, R, of the signal is defined as

$$R = \{x[n]\}_{\max} - \{x[n]\}_{\min}$$

- The quantisation step size or resolution is defined as $\Delta V = \frac{R}{L}$ where L is the number of quantisation levels ($L = 2^B$)

$$\therefore SQNR(dB) = 10\,log_{10}\frac{P_s}{(R/L)^2/12} = 10\,log_{10}\frac{12P_sL^2}{R^2}$$

$$SQNR(dB) = 10\,log_{10}(P_s) + 20\,log_{10}(L) + 10\,log_{10}(1\,2) - 20\log(R)$$

Signal-to-Quantisation Noise Ratio (SQNR)

SQNR for a Sine Wave

- For the sine wave input, the average signal power is $A^2/2$, since the RMS value is $A/\sqrt{2}$. The signal-to-quantisation noise power ratio (SQNR) in decibels is:

$$SQNR = 10\,log\left(\frac{\frac{A^2}{2}}{\frac{\Delta V^2}{12}}\right) = 10\,log\left(\frac{\frac{A^2}{2}}{\frac{(2A/2^B)^2}{12}}\right) = 10\,log\left(\frac{3\times 2^{2B}}{2}\right)$$

$$\boldsymbol{SQNR = 6.02B + 1.76 \quad dB}$$

- The SQNR increases with the number of bits, *B*. In many DSP applications, an A/D converter resolution of 12 to 16 bits is adequate.
- It can be seen from the table below that the signal-to-quantisation noise ratio increases by approximately 6dB for each additional bit.

Number of bits	Number of Levels	SQNR
3	8	19.7 dB
4	16	25.3 dB
5	32	31.6 dB
6	64	37.7 dB
7	128	43.8 dB

Example 18:

- A sampled signal that varies between -2V and + 2V is quantised using B bits. What value of B will ensure that the quantisation noise power is less than 25×10^{-6}?

$$\text{Resolution } (\Delta V) = \frac{R \text{ (Dynamic range)}}{L \text{ (No of levels in the quantiser)}}$$

$$\sigma_e^2 = \frac{\Delta V^2}{12} < 25 \times 10^{-6};\ \therefore\ \Delta V^2 < 12 \text{ x } 25 \times 10^{-6}$$

R = 2 – (-2) =4; $\therefore\ \Delta V^2 = \left(\frac{R}{L}\right)^2 = \frac{16}{L^2}$

$\frac{16}{L^2} < 12 \text{ x } 25 \times 10^{-6}$; L > 230.94

L = 2^B ; $\therefore\ 2^B > 230.94$

B > 7.85;

Therefore, the minimum number of bits required is B = 8 ;

Digital to Analogue Conversion (D/A)

D/A converter

- The D/A conversion process converts a digitally processed signal back into analogue form.
- This is required, for example, to generate an analogue audio signal to drive a loudspeaker or to produce an alarm signal. The D/A process is shown in Figure 16.
- The digital input y[n] consists of a sequence of numerical sample values. The D/A converter converts each digital value into a corresponding analogue voltage or current.
- In a practical D/A converter, each output value is held constant for one sampling period T, until the next digital sample arrives.
- This produces a **staircase-shaped waveform**, as shown in Figure 16.
- The D/A shown in Figure 16 is referred to as a **zero-order hold** (i.e. holding each sample value for one sample interval)

$$\text{Digital samples } y[n] \rightarrow \text{D/A} + \text{hold} \rightarrow \hat{y}(t)$$

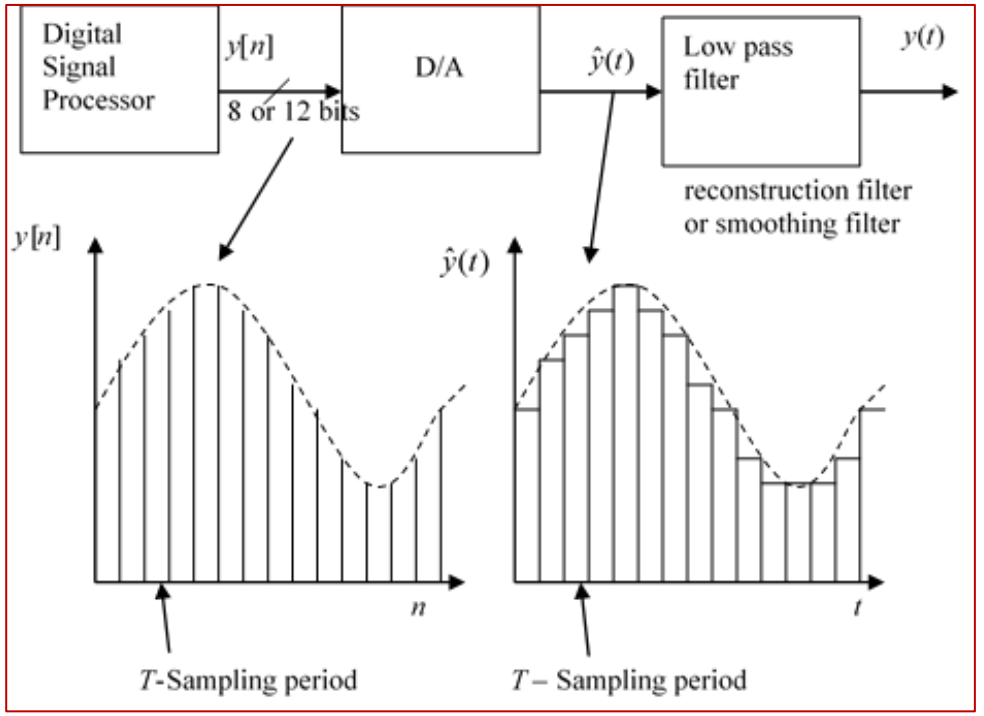


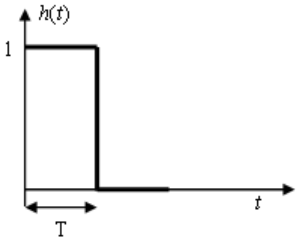
Figure 16: Conversion process from digital samples to analogue signals.

D/A Converter: Zero-Order Hold

Zero- Order Hold

- In a zero-order hold D/A converter, each sample value is held constant for one sampling period T.
- Therefore, the D/A output can be viewed as a sequence of **rectangular pulses**, each of duration T.
- Consider one rectangular pulse:

$$h(t) = \begin{cases} 1 & 0 \le t \le T \\ 0 & otherwise \end{cases}$$



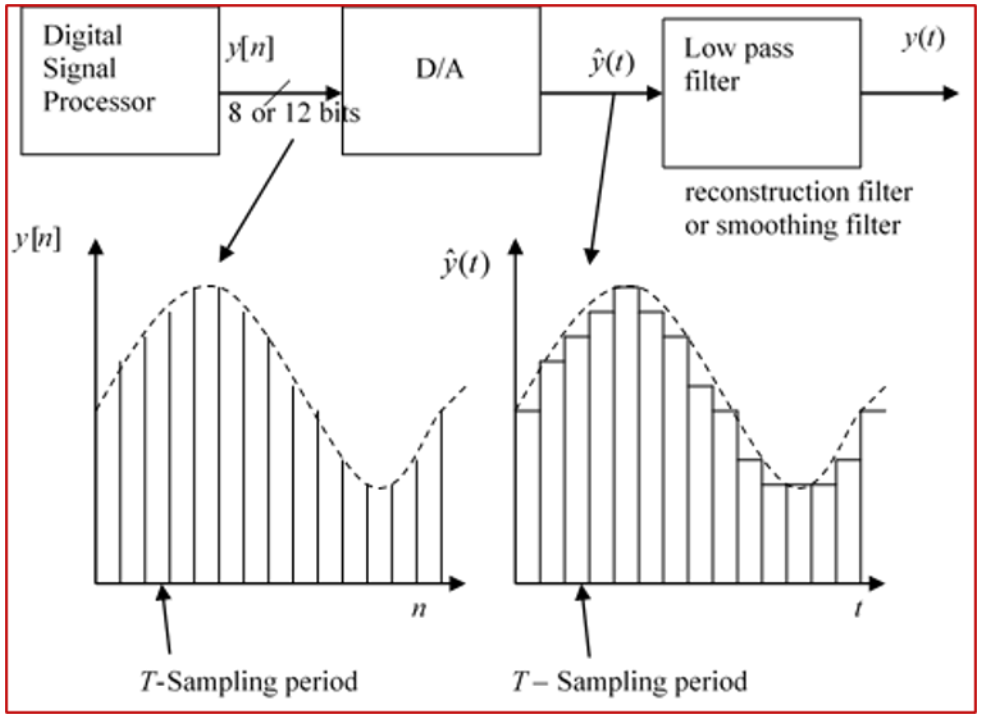
- The frequency response of the zero-order hold is:

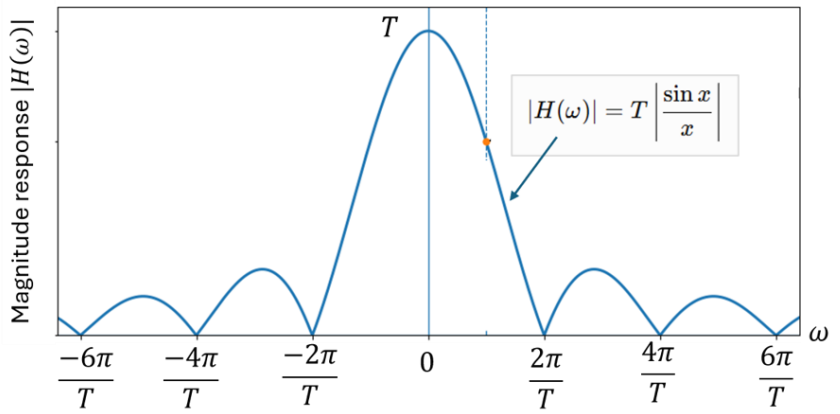
$$H(\omega) = Te^{-j\omega T/2}\frac{\sin(\omega T/2)}{\omega T/2}$$

- Therefore, its magnitude response is

$$|H(\omega)| = T\left|\frac{\sin(\omega T/2)}{\omega T/2}\right| \qquad |H(\omega)| = T\left|\frac{\sin x}{x}\right| \qquad x = \frac{\omega T}{2}$$

- The zero-order hold does not have a flat frequency response — its amplitude decreases as frequency increases.





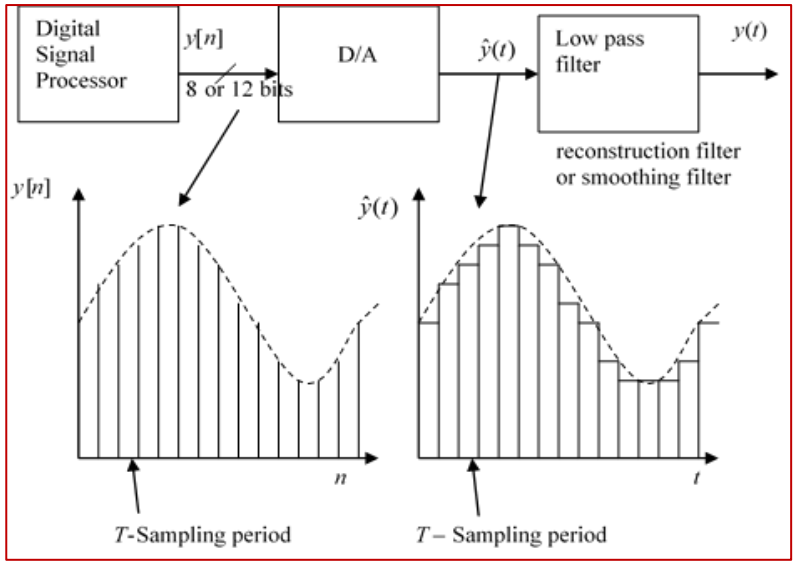
$Sin\ x/x$ Distortion and Reconstruction Filter

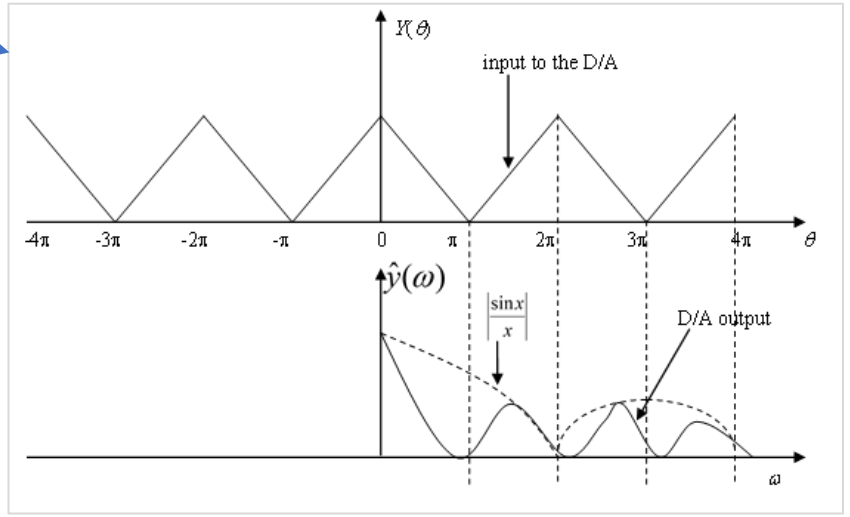
Effect of the zero-Order Hold

- The zero-order hold causes the D/A output spectrum to be shaped by the $\sin x/x$ response.
- As frequency increases, the sinx/x response decreases. Therefore, higher-frequency components are attenuated more than lower-frequency components.
- This produces **amplitude distortion**, sometimes called **aperture distortion**.

$$\text{Sampled spectrum} \times \underbrace{\frac{\sin x}{x}}_{\text{ZOH response}} \longrightarrow \text{D/A output spectrum}$$

- The amplitude of the output signal spectrum is multiplied by the $\sin x/x$ function, which acts like a low-pass filter, with the high frequencies heavily attenuated.
- The $\sin x/x$ effect is due to the holding action of the D/A converter and, in signal recovery, introduces an **amplitude distortion**
- The D/A output also contains unwanted spectral images centred at multiples of the sampling frequency f_s.
- A **low-pass reconstruction filter** removes these unwanted high-frequency images and produces a smooth analogue output signal.





Topic 1 - Summary

Summary

- How to distinguish between continuous-time (analog) and discrete-time (digital) signals.
- How to distinguish between differential equations and difference equations.
- Manipulation of discrete-time signals including the unit impulse and its properties.
- The definition of the Continuous-Time Fourier Transform (FT) and its Inverse.
- Continuous-time and discrete-time periodic signals and their definitions.
- The relationship between analogue and digital frequency: $\theta = \frac{2\pi f_a}{f_s}$; $N = \frac{2\pi k}{\theta} = \frac{k f_s}{f_a}$.

  where *N* is the number of samples in a period; θ is the digital frequency with $-\pi \leq \theta \leq \pi$
- Sampling of analog signals; the sampling theorem, Nyquist frequency and limitations such as aliasing.
- Sampling of bandpass signal.
- Calculations of aliased frequencies: $\boldsymbol{f'_n = f_n - kfs}$ $(-\frac{f_s}{2} \leq \boldsymbol{f'_n} \leq \frac{f_s}{2})$, $\boldsymbol{f'_n}$ is the aliased frequency; $\boldsymbol{f_n}$ is the original frequency causing the aliasing and *k* is an integer .
- How to demonstrate the effect of aliasing using sketched magnitude spectrum plots.
- A block diagram of the conversion from analog to digital and back to analog form, including the function of each block.
- Analogue-to-digital conversion, particularly amplitude quantisation and quantisation error. Be able to calculate the signal-to-quantisation noise ratio (SQNR).
- Digital-to-analogue conversion and the role of the zero-order hold and reconstruction filter. Show your understanding using both mathematical and hand-sketched explanations.